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IS IT POSSIBLE TO CREATE A UNIVERSE IN THE LABORATORY BY QUANTUM TUNNELING?

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ABSTRACT

We explore the possibility that a new universe can be created by producing a small bubble of false vacuum. The initial bubble is small enough to be produced without an initial singularity, but classically it could not become a universe— instead it would reach a maximum radius and then collapse. We investigate the possibility that quantum effects allow the bubble to tunnel into a larger bubble, of the same mass, which would then classically evolve to become a new universe. The calculation of the tunneling amplitude is attempted, in lowest order semiclassical approximation (in the thin-wall limit), using both a canonical and a functional integral approach. The canonical approach is found to have flaws, attributable to our method of spacetime slicing. The functional integral approach leads to a Euclidean interpolating solution that is not a manifold. To describe it, we define an object which we call a "pseudomanifold", and give a prescription to define its action. We conjecture that the tunneling probability to produce a new universe can be approximated using this action, and we show that this leads to a plausible result.

I. INTRODUCTION

In this paper we address the question of whether the laws of physics, as we know them, allow for the possibility that human beings could create a new universe.¹

This question may at first seem absurd, and it *would* be absurd to ask this question in the context of the standard big bang model. In that picture the visible universe began, say at $t \approx 1$ sec, with about 10^{32} solar masses of material. Inflationary cosmology, on the other hand, gives a radically different answer to the question of what it takes to produce a universe. For inflation at a typical grand unified theory scale of 10^{14} GeV, for example, the universe emerges at $t \approx 10^{-35}$ sec from a region of false vacuum with a radius of only $\sim 10^{-24}$ cm and a mass of only ~ 10 kg. This list of ingredients is so modest that the notion of a human-made universe can be seriously entertained.

The achievement of a mass density of $\sim 10^{76}$ g/cm³ is admittedly somewhat beyond our technology— it is in fact well beyond the range of any foreseeable technology. For purposes of discussion, however, we will assume that some civilization in the distant future may have the necessary capabilities. We will focus only on the question of principle, the question of whether the possibility of creating a new universe is excluded by the laws of physics.

While the question is interesting in its own right, it is also a useful device to focus attention on the minimal conditions necessary for a universe to be created, either in the

laboratory or in the cosmos. In addition, the problem is a very interesting test of our understanding of quantum gravity. Since there is no well-formulated theory of quantum gravity it is useful to have specific models in which calculations can be attempted. We look at a semiclassical approximation to quantum gravity in two ways: canonically and through the functional integral. In both cases we encounter results which we do not fully understand. The functional integral approach, however, gives a result that we feel is plausible, and likely to be correct.

Specifically, we study the possibility that a new universe can be created by producing a spherically symmetric bubble of false vacuum (*i.e.*, a metastable state in which a scalar field has an expectation value that differs from its vacuum value), which will then undergo inflation. After the bubble has settled into the true ground state phase, the new universe would evolve much like our own universe is believed to have evolved after its period of inflationary growth. For simplicity, we will assume throughout the paper that the wall of the bubble is small compared to any other relevant dimension (thin-wall approximation), so that it can be described by a single coordinate, its radius.

We will not discuss the details of how a false vacuum bubble might be produced, except to mention here that there are at least two possible mechanisms: one involves supercooling, and the other involves the compression of fermions. The supercooling mechanism would work in a particle theory in which at high temperatures the scalar field associated with the false vacuum undergoes a phase transition, acquiring an expectation value at or near its expectation value in the false vacuum. If the false vacuum is sufficiently stable, then the scalar field could be trapped in the false vacuum state if a region of space in the high temperature phase was cooled rapidly enough. The fermion compression mechanism would work in a particle theory in which the vacuum expectation value of the scalar field gives rise to a fermion mass. A high density of fermions would then exert a "force" on the scalar field expectation value, driving it near the false vacuum value where the fermion mass is reduced. Due to the presence of the fermions, however, such a state is not yet a false vacuum—it would in fact have a positive pressure. A false vacuum, with its characteristically negative pressure, could be obtained if the fermions were rapidly swept away.

Classically, *i.e.* in the absence of quantum mechanics, a variety of authors²⁻⁷ have solved Einstein's equations for a spherical region of false vacuum sitting in an asymptotically flat empty space. (Our discussion follows most closely that of Ref. [6].) The true vacuum region is described by the Schwarzschild geometry, with a mass M which is by definition equal to the total energy of the bubble. Inside the bubble the geometry is de Sitter space, with a Hubble constant χ given by $\chi^2 = 8\pi G\rho_f/3$, where ρ_f is the false vacuum energy density. The false vacuum is separated from the true vacuum by a thin wall with energy per unit area σ . (ρ_f and σ are taken as fixed constants.) Einstein's equations at the wall imply a set of junction conditions,⁸ which in turn lead to the equation of motion for the wall.

The behavior of the bubble wall trajectory depends on the mass M of the bubble. If M is larger than a critical mass M_{cr} (which depends on ρ_f and σ), then the radius of the bubble wall (and, accordingly, the volume of the bubble) starts from zero and then grows without bound. The bubble does not, however, expand into the external true vacuum region. In fact, from the point of view of an external observer, the expanding bubble always remains

behind the Schwarzschild horizon. A bubble of this type would produce a new universe, but one must also ask whether a bubble of this type can be produced in the laboratory.

The exact solution referred to above has the undesirable feature of beginning with an initial singularity. An initial singularity is a point in spacetime at which at least one backward-going null geodesic terminates. Such a geodesic has no history whatever prior to the singularity, and therefore the presence or absence of a light ray traveling along the geodesic cannot be determined by initial conditions set prior to the singularity. For the exact solutions, the singularity takes the form of an initial spacelike boundary of the manifold, marking the terminal point for any backward-going timelike or null trajectory that intersects it. The singularity in this case is also a place of infinite gravitational force, and clearly represents a singular initial value configuration. We therefore conclude that any plausible scheme to create a universe in the laboratory must avoid an initial singularity.

Thus, we are led to ask whether it is possible for a system with nonsingular initial data to evolve so that its late-time behavior could closely resemble the late-time behavior of the exact ($M > M_{cr}$) solutions discussed above. As an analogy, note that an eternal black hole begins with an initial singularity, but that a black hole solution with very similar late-time behavior can be obtained from the nonsingular initial conditions of stellar collapse. For the false vacuum bubble, however, the initial singularity appears to be essential, at least at the level of classical physics. Making certain weak assumptions, it has been shown⁹ that if the bubble radius grows large (greater than $1/\chi$), then the Penrose theorem¹⁰ implies that the spacetime must have emerged from an initial singularity. The proof assumes that the spacetime geometry is governed fully by classical general relativity.

In this paper we will discuss whether this obstacle might be overcome by quantum tunneling. We will consider bubbles of mass $M < M_{cr}$; for any such M there are two types of solutions, which we call types (a) and (b). For solutions of type (a), the radius starts at zero, reaches a maximum value, and then decreases again to zero. These solutions are influenced mainly by the pressure gradient, which points inward from the zero pressure true vacuum toward the negative pressure false vacuum. A solution of type (b) can be described roughly, at a fixed time, as more than half of a closed universe, connected to the asymptotic space by a neck. This solution begins at an asymptotically large radius in the asymptotic past, shrinks to a minimum radius, and then grows again without bound. At large times this solution closely resembles the solutions for $M > M_{cr}$ that were discussed earlier, and would grow to become a universe. The Penrose theorem, however, implies that these solutions must also begin from a singular spacetime. For solutions of type (a), the Penrose theorem may or may not apply, depending on the mass and on the stage of evolution of the bubble. We are primarily interested in creating a bubble during its period of expansion. We will show in Sec. II that the Penrose theorem applies to an expanding type (a) solution if and only if $M > M_S$, where M_S is the mass of a (type (a)) solution for which the maximum radius is precisely equal to the Schwarzschild radius. (M_S depends on ρ_f and σ , and in all cases $M_S < M_{cr}$.)

We therefore consider an initial false vacuum bubble with $M < M_S$, which can in principle be created without an initial singularity. (Note that as M gets smaller and smaller, the role of gravity becomes increasingly irrelevant. Thus the creation of the bubble reduces to a problem involving only quantum field theory in flat space.) While classically such a

bubble would reach a maximum radius and then contract, we would expect that there is a nonzero probability that it might instead tunnel to a solution of type (b)—the two states have the same energy, and are not distinguished by any other conserved quantity. From the outside, the bubble would appear to have vanished, leaving behind a black hole of the same mass. On the inside, the growing universe would again be susceptible to a classical description, just as a particle can be described classically after it has tunneled through a barrier and is again in a classically allowed region. The new universe would grow large, but the classical argument implying an initial singularity would be invalid, since the spacetime took a quantum leap during which it was not governed by Einstein's equations.

In this paper we attempt to calculate the tunneling probability, using a semiclassical approximation to quantum gravity. However before quantizing a system it is important to understand its classical motions. In the next section we review the salient features of the classical trajectories of the thin-wall bubble problem. In particular we focus on the application of the singularity theorem, identifying those cases where the singularities are a necessary part of the spacetime. We then discuss the classical action whose variation gives rise to the classical equations of motion. Knowledge of the classical action, and its associated Lagrangian, is a prerequisite to both the canonical and functional integral quantization methods. In Sec. IV we review the semiclassical formalism for quantum tunneling, showing that the amplitude is determined to lowest order in $\hbar$ by the action of a Euclidean solution that interpolates between the initial and final classical states; in the course of this calculation we find a formula for the quantity that must be subtracted from the action to properly define its zero point. In Sec. V we estimate the tunneling rate using a functional integral approach. Here we assume that the functional integral is dominated by a configuration which is a solution to the Euclidean field equations. In applying this technique to the thin-wall bubble problem, we calculate the imaginary-time solution for the trajectory of the bubble wall. We find, however, that it turns back on itself in a way that does not allow the construction of an interpolating four-dimensional manifold. Nonetheless we present a prescription for evaluating the subtracted action for this interpolating object, and we obtain what we feel is a sensible answer. This is the main result of our paper. In Sec. VI we discuss the purely canonical approach to the thin-wall bubble problem. This approach founders almost immediately, but we carry through the calculations just to see what we get. In contrast to the functional integral result, the answer in this case does not seem sensible. Finally in Sec. VII we summarize our conclusions.

The quantum behavior of false vacuum bubbles has also been discussed in several earlier papers. Frolov, Markov, and Mukhanov¹¹ have conjectured that quantum effects give rise to a limiting spacetime curvature, and in this context they have discussed a scenario in which black hole collapse leads to the production of a new universe. Berezin, Kuzmin, and Tkachev¹² have discussed the tunneling of false vacuum bubbles, using Euclidean semiclassical methods similar to the functional integral approach we present in Sec. V. They assumed, however, that the Euclidean bubble wall trajectory forms a closed loop, and they therefore obtained results very different from ours. Berezin, Kozimirov, Kuzmin, and Tkachev¹³ have developed a quantum theory of thin-wall bubbles, constructing a Hamiltonian by using the classical equations of motion and the criterion that it should numerically equal the mass of the false vacuum bubble. Their Hamiltonian is different from the one that we find, however,

because they assumed that the Hamiltonian contains no explicit dependence on the mass. Their formalism seems to also have the shortcoming that the quantities σ_{in} and σ_{out} , corresponding to the signs of the functions β_S and β_D that we introduce in Sec. II, are treated as parameters and not dynamical variables. Thus, the formalism apparently cannot describe motion for which the σ 's change sign, whether the motion is classically allowed or classically forbidden. This would prevent, for example, a description of tunneling for false vacuum bubbles with mass less than M_S . In addition, Goncharov, Linde, and Mukhanov¹⁴ have used stochastic methods to estimate the rate for the nucleation of false vacuum bubbles in empty Minkowski space. Unlike semiclassical techniques which attempt to calculate the exponent B of the tunneling amplitude $Ae^{-B/\hbar}$, the stochastic approach attempts only to estimate the magnitude of B . Finally, as we were completing this work we received a preprint by Fischler, Morgan, and Polchinski¹⁵ which addresses exactly the same question addressed here. We will discuss the results of this paper in our conclusion, Sec. VII.

II. THE CLASSICAL TRAJECTORIES

The starting point for our discussion is the classical problem of a spherical region of false vacuum, with energy per unit volume ρ_f , separated from empty space by a thin wall with energy per unit area σ . The goal is to describe the full spacetime geometry, including the motion of the wall. This problem has been analyzed previously²⁻⁷, so here we will only state the results, without derivations. We follow the notation of Ref. [6]. An alternative derivation of the equation of motion, using a variational principle, will be presented in the next section.

The empty-space part of the solution is the Schwarzschild geometry. The metric in Schwarzschild coordinates is given by:

$$ds^2 = -A_S dT_S^2 + A_S^{-1} dR^2 + R^2 d\Omega^2, \quad (2.1)$$

where

$$A_S = 1 - \frac{2GM}{R}, \quad (2.2)$$

$d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$, and M is the total energy of the bubble. Since these coordinates do not cover all of Schwarzschild space and since they have a coordinate singularity at $R = 2GM$, it is useful to introduce Kruskal-Szekeres¹⁶ coordinates (V_S, U_S) : these coordinates cover the entire manifold and have no coordinate singularities. They also have the convenient property that lightlike lines travel at 45° . The metric can then be written as

$$ds^2 = 32 \frac{(GM)^3}{R} \exp\left(-\frac{R}{2GM}\right) (-dV_S^2 + dU_S^2) + R^2 d\Omega^2, \quad (2.3)$$

where

$$\left(\frac{R}{2GM} - 1\right) \exp\left(\frac{R}{2GM}\right) = U_S^2 - V_S^2. \quad (2.4)$$

Both the Schwarzschild and Kruskal-Szekeres metrics contain the term $R^2 d\Omega^2$, implying that R has the same meaning in both systems. Lines of constant T_S are straight lines through the origin of the (V_S, U_S) plane, with

$$T_S = \begin{cases} 4GM \tanh^{-1}(V_S/U_S) & \text{if } |V_S/U_S| < 1 \\ 4GM \tanh^{-1}(U_S/V_S) & \text{if } |V_S/U_S| > 1. \end{cases} \quad (2.5)$$

The relation between the two coordinate systems is described more fully in Ref. [6].

Inside the bubble the spacetime geometry is de Sitter space. The metric can be written in static coordinates as:

$$ds^2 = -A_D dT_D^2 + A_D^{-1} dR^2 + R^2 d\Omega^2, \quad (2.6)$$

where

$$A_D = 1 - \chi^2 R^2. \quad (2.7)$$

These coordinates have a coordinate singularity at $R = 1/\chi$, and they fail to cover the entire de Sitter manifold. Both of these shortcomings can be avoided by using the Gibbons-Hawking¹⁷ coordinates (V_D, U_D) , in terms of which the metric is

$$ds^2 = \chi^{-2} (1 + \chi R)^2 (-dV_D^2 + dU_D^2) + R^2 d\Omega^2, \quad (2.8)$$

where

$$\frac{1 - \chi R}{1 + \chi R} = U_D^2 - V_D^2. \quad (2.9)$$

Again lightlike lines travel at 45° , and the presence of the term $R^2 d\Omega^2$ implies that R has the same meaning in all of the coordinate systems that we have introduced. Lines of constant T_D are again straight lines through the origin, with

$$T_D = \begin{cases} \chi^{-1} \tanh^{-1}(V_D/U_D) & \text{if } |V_D/U_D| < 1 \\ \chi^{-1} \tanh^{-1}(U_D/V_D) & \text{if } |V_D/U_D| > 1. \end{cases} \quad (2.10)$$

The relation between the two de Sitter coordinate systems is also described more fully in Ref. [6].

The use of the Schwarzschild and de Sitter metrics guarantees that the Einstein equations are satisfied both outside and inside the bubble. The equation of motion for the bubble wall is then obtained by insisting that the Einstein equations hold also at the boundary. The equation of motion can be expressed in terms of the function $r(\tau)$, where r is the value of the coordinate R at the wall, and τ is the proper time as measured by an observer who rides along with the wall. We will also use the notation $t_S(\tau)$ to denote the value of the Schwarzschild time variable T_S on the wall, and $t_D(\tau)$ to denote the value of T_D on the wall. The motion of the wall is determined by

$$\beta_D - \beta_S = \kappa r, \quad (2.11)$$

where¹⁸

$$\begin{aligned} \beta_S &\equiv A_S \dot{t}_S, \\ \beta_D &\equiv -A_D \dot{t}_D, \end{aligned} \quad (2.12)$$

and

$$\kappa \equiv 4\pi G\sigma, \quad (2.13)$$

where the overdot ($\dot{}$) denotes a derivative with respect to τ . By requiring the velocity four-vector to be normalized, one finds

$$\begin{aligned} \beta_D &= \pm \sqrt{A_D + \dot{r}^2}, \\ \beta_S &= \pm \sqrt{A_S + \dot{r}^2}, \end{aligned} \quad (2.14)$$

so Eqs. (2.11) and (2.14) form a closed set of equations that can be solved to give $r(\tau)$. Note that Eqs. (2.12) imply that β_S is positive (negative) if the polar angle $\tan^{-1}(V_S/U_S)$ is increasing (decreasing) along the trajectory, while β_D is positive (negative) if the polar angle $\tan^{-1}(V_D/U_D)$ is decreasing (increasing) along the trajectory.

We can understand the qualitative features of the solutions to Eqs. (2.11) and (2.14) by rewriting them as

$$\dot{r}^2 + V(r, M) = -1, \quad (2.15)$$

which describes one-dimensional motion with an “energy” equal to -1 , and a “potential-energy” function given by

$$V(r, M) = -(2GM)^{2/3} (\chi^2 + \kappa^2)^{1/3} \left\{ \frac{(1 - z^3)^2}{\gamma^2 z^4} + \frac{1}{z} \right\}. \quad (2.16)$$

Here

$$z^3 \equiv \frac{\chi^2 + \kappa^2}{2GM} r^3, \quad (2.17)$$

and

$$\gamma^2 \equiv \frac{4\kappa^2}{\chi^2 + \kappa^2}. \quad (2.18)$$

The rescaled radius coordinate z has been introduced to make it easier to see the effect of varying the mass M .

A plot of $V(r, M)$ is shown in Fig. 1, for a variety of values of M . Note that the dependence on M can be described by shifting the curve vertically — as M is increased, the curve is shifted downward. The graph at the upper left shows the form of the curve for the largest values of M , for which the entire curve is shifted below -1 . For this range of masses r varies monotonically with τ .

Fig. 1 also shows the values of z for which β_S and β_D change sign. From Eqs. (2.11) and (2.14), one can show that β_S changes sign at $z = 1$, while β_D changes sign at $z \equiv z_D = (1 - \frac{1}{2}\gamma^2)^{-1/3}$. The maximum of the curve, z_m , occurs at a value of z between 1 and z_D .

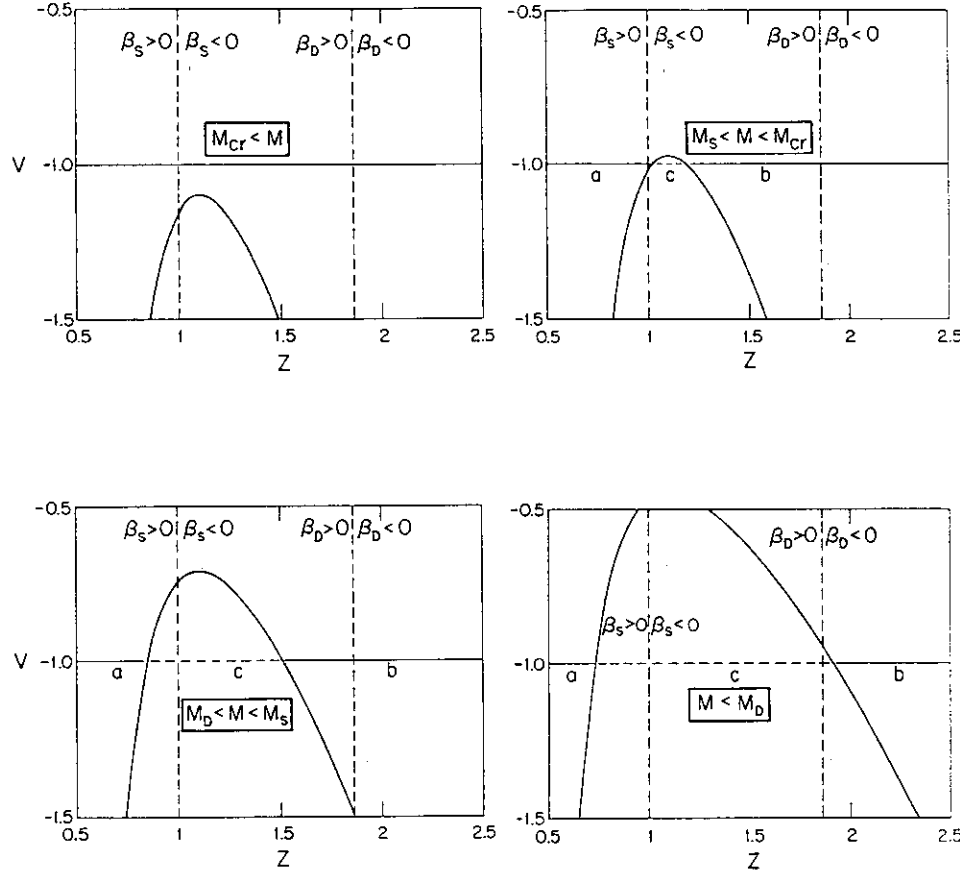


Fig. 1: Graph of the potential-energy function $V(z, M)$, using the rescaled radius variable z , shown for various ranges of the bubble mass M . The bubble wall trajectory $r(r)$ is identical to the motion of a particle in this potential, with energy $E = -1$. For $M < M_{cr}$, there are two classically allowed trajectories, labeled (a) and (b). The diagram also shows the path of a tunneling trajectory, shown as a dashed line and labeled (c). The figure shows the line $z = 1$ at which β_S changes sign, and the line $z = z_D$ at which β_D changes sign. (The four graphs are plotted for $\gamma = 1.3$, with M/M_{cr} taken as 1.15, 0.965, 0.6, and 0.3, respectively.)

The behavior of the trajectory depends on the mass M , and it is useful to define three quantities, M_{cr} , M_S , and M_D , by the relations

$$\begin{aligned} V(z_m, M_{cr}) &\equiv -1 \\ V(1, M_S) &\equiv -1 \\ V(z_D, M_D) &\equiv -1. \end{aligned} \quad (2.19)$$

It can be shown that $M_D < M_S < M_{cr}$.

For any $M > M_{cr}$, $V(r, M)$ will resemble the upper left graph, with monotonic trajectories. As M is decreased below M_{cr} , but still above M_S , the curve has the form shown in the upper right. Now the curve shows a potential barrier, with a classically allowed region at small r corresponding to a trajectory of type (a), and a classically allowed region at large r corresponding to a trajectory of type (b). Both trajectories have the same mass M , and eventually we will be interested in tunneling between them. For later reference the tunneling trajectory is shown here as (c). Note that β_S changes sign along the type (a) trajectory, while β_D changes sign along the type (b) trajectory.

Now consider decreasing M until it reaches M_S . In that case $\beta_S = 0$ at the endpoint of the trajectory, where $\dot{r} = 0$; therefore $A_S = 0$ at this point, implying that $r = 2GM$. Thus, M_S is the mass for which a type (a) trajectory just barely reaches its Schwarzschild horizon. Similarly, it can be seen that M_D is the mass for which a type (b) trajectory, at its minimum radius, just barely reaches the de Sitter horizon.

The curve for $M_D < M < M_S$ is shown in the lower left of Fig. 1. Note that in this case β_S is uniformly positive for trajectories of type (a), while β_D changes sign along trajectories of type (b). The graph at the lower right shows the curve for $M < M_D$, for which case neither β_S nor β_D change sign in the classically allowed regions.

Given a solution $r(\tau)$, we would still like to obtain the full spacetime trajectory, which can be described by specifying a path $(V_S(\tau), U_S(\tau))$ as seen from the Schwarzschild side, and a path $(V_D(\tau), U_D(\tau))$ as seen from the de Sitter side. (There is no reason for these coordinates to agree at the wall, but $r(\tau)$ must have the same value when calculated from either coordinate system.) The additional information is contained in Eqs. (2.12), allowing for a full determination of the trajectories. The details are worked out in Appendix C of Ref. [6]. Here we will review the qualitative behavior of the solutions, and we will also summarize the results of Ref. [9] concerning the implications of the Penrose theorem regarding initial singularities.

Fig. 2 shows the behavior of a solution with $M > M_{cr}$. As described in Sec. I, the radius of this solution grows without bound, but the growth takes place entirely in the left quadrant of the Kruskal diagram. The bubble wall never crosses the Schwarzschild horizon to enter the external region in the right quadrant. The beginning of the new universe is shown in the upper left hand corner of the de Sitter diagram.

The point labeled P indicates an example of a closed "anti-trapped" surface, a necessary condition for the Penrose theorem. (Recall that angular variables are suppressed, so that a point in the diagram actually corresponds to a closed surface.) A closed surface is called "anti-trapped" if it has the property that both the inward- and outward-going sets of light

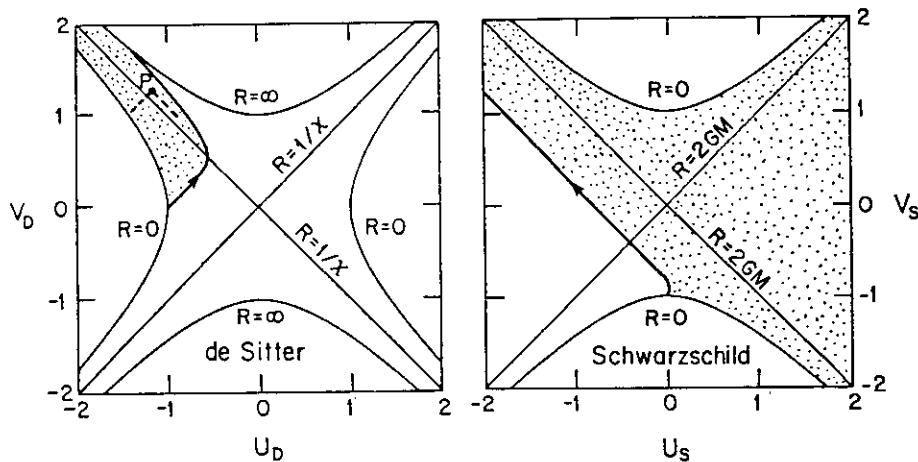


Fig. 2: The spacetime trajectory for a solution with $M > M_{cr}$. Here the trajectories are shown on a full de Sitter space coordinate system and on a full Schwarzschild space coordinate system. The actual bubble spacetime, however, consists only of the de Sitter region to the left of the wall trajectory and the Schwarzschild region to the right of the wall trajectory, as indicated by the dotted areas of the diagram. The two spaces are sewn together along the wall, indicated by a heavy line with an arrow on it. The point P shows an example of a closed "anti-trapped" surface, indicating that the Penrose theorem applies. The light rays arriving at P are shown as dashed lines. (The figure is drawn for $\gamma = 1.3$ and $M/M_{cr} = 3$.)

rays, arriving normally at the surface, are diverging at every point on the surface. In the figure, the outward-going light rays arriving at P are shown as a dashed line emanating from the $R = 0$ boundary at the left of the de Sitter diagram. Although the divergence of the light rays at P is determined completely by the local geometry at P , it is nonetheless very convenient to use the global properties of de Sitter space to find the sign of the divergence. From Eq. (2.9) we see that R varies monotonically along a light ray in de Sitter space, so R must be increasing as the rays reach P —thus the rays are diverging. The inward-going light rays arriving at P are shown as a dashed line, which, if it were continued, would extend back to the $R = 0$ boundary at the right of the de Sitter diagram. These light rays are therefore also diverging, so the surface P is anti-trapped. The Penrose theorem then implies that if there exists a noncompact Cauchy surface (which there does), and if the energy-momentum tensor satisfies $T_{\mu\nu}k^\mu k^\nu \geq 0$ for all null vectors k^μ , then there exists an initial singularity. Thus, at the level of classical physics, there is no way that one can modify the early part of this diagram to produce the same result without an initial singularity.

As the mass is lowered below M_{cr} , the barrier appears in the potential $V(r, M)$. For M below M_{cr} but still above M_S , the type (a) solution has the form shown in Fig. 3. The point indicated by P is a closed anti-trapped surface, as can be seen by tracing back the light rays arriving at P , shown as dashed lines. Thus the Penrose theorem implies that an initial singularity is necessary to classically produce this false vacuum bubble at any time earlier than P . Since P is the point of maximum radius r , we conclude that classical physics does not permit the creation of this bubble in its expanding phase.¹⁹ Note, however,

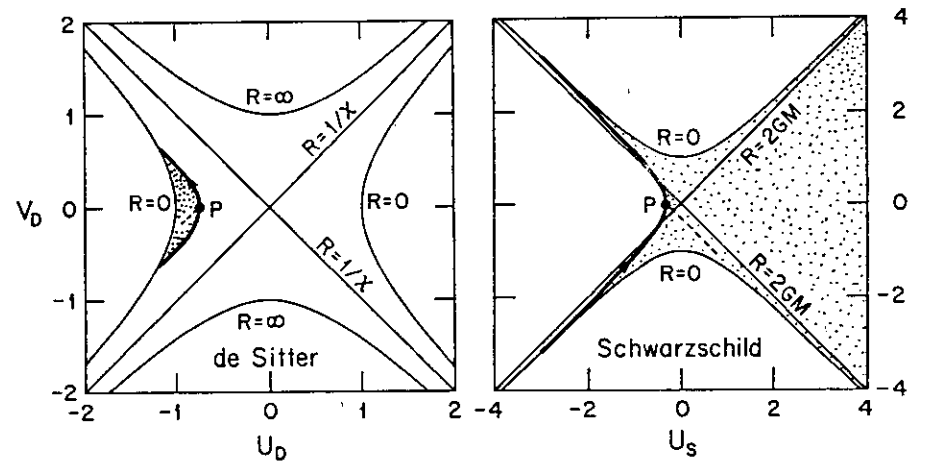


Fig. 3: The spacetime trajectory for a solution of type (a), with $M_S < M < M_{cr}$. The coordinates have been chosen so that $\dot{U}_S$ and $\dot{U}_D$ are zero when $\dot{r}$ is zero. The point P is again used to denote the location of a closed anti-trapped surface, and the light rays arriving at P are shown as dashed lines. (The figure is drawn for $\gamma = 1.95$ and $M/M_{cr} = .97$.)

that the closed anti-trapped surface at the point of maximum radius would not exist if the trajectory on the Kruskal diagram crossed to the right of the origin, which is what happens for $M < M_S$.

Thus, an expanding false vacuum bubbles with $M < M_S$ can in principle be produced classically without an initial singularity. Since these bubbles are in principle experimentally accessible, we will be interested in them as a starting point for the tunneling calculation.

Figs. 4 and 5 show the type (a) and type (b) solutions, respectively, for a low value of M , a value that is below M_D . The type (a) solution of Fig. 4 will form the starting point of the tunneling calculation. Although the diagram shows an initial singularity, in this case the Penrose theorem does not imply that the singularity is required. In fact, gravity becomes increasingly irrelevant as M is made arbitrarily small, so the problem of producing a bubble of this type reduces to a problem in flat space. We can thus imagine cutting the diagram along a spacelike hypersurface Σ at some point after the bubble wall has emerged beyond the Schwarzschild horizon, but before it reaches its maximum radius. The state of the bubble on Σ can then presumably be produced from nonsingular initial conditions. That is, we assume that it is possible in principle to design machinery that would create, from nonsingular initial conditions, a state in which the value of the scalar field and its time derivative in the vicinity of the bubble would agree with the values on Σ . (Elsewhere the state created by the machinery would of course be very different, and for example would include the machinery itself.) While classically this type (a) solution reaches a maximum radius and then collapses to a singularity, we will be asking whether quantum theory allows it to tunnel to the solution of type (b). Note that the solution of type (b) contains closed anti-trapped surfaces, indicating that classically such a configuration cannot be created without an initial singularity. If it can appear, however, by quantum tunneling, then the requirement of an

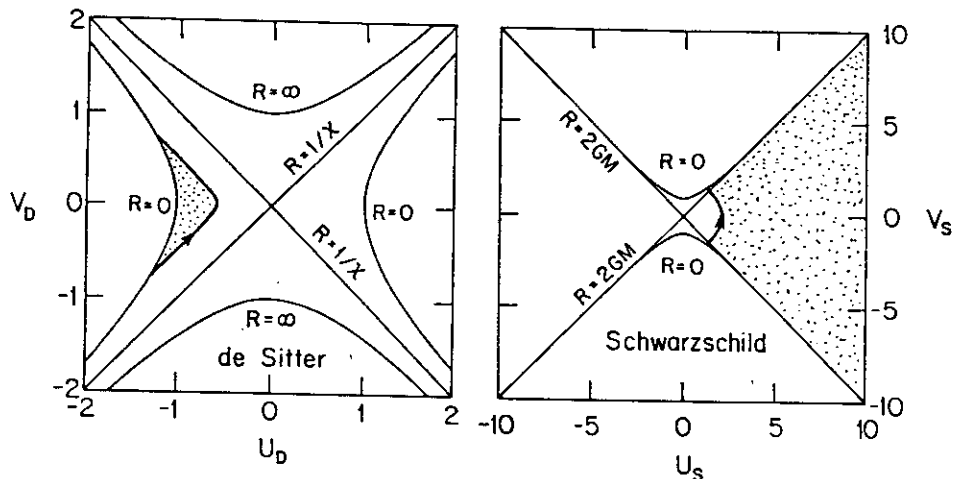


Fig. 4: The spacetime trajectory for a solution of type (a), with $M < M_D$. This solution will be the initial state for the tunneling calculation. (The figure is drawn for $\gamma = .75$ and $M/M_{cr} = .3$.)

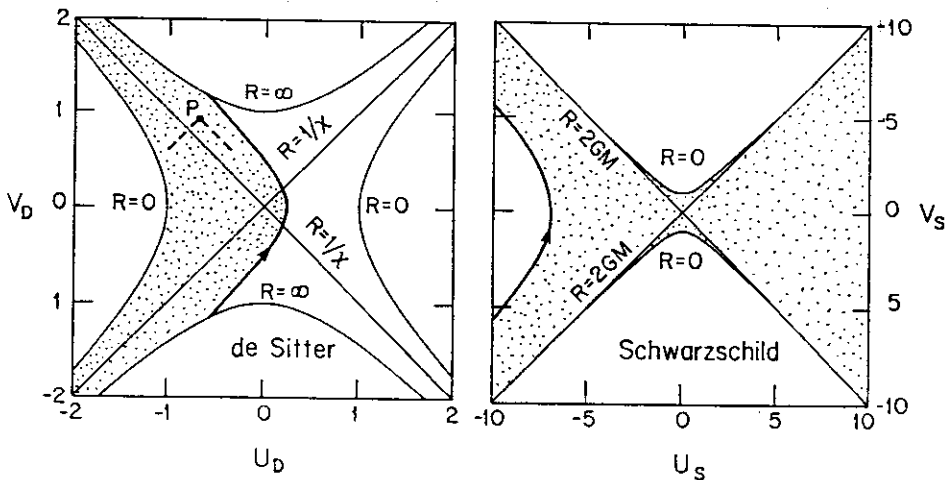


Fig. 5: The spacetime trajectory for a solution of type (b), with $M < M_D$. This solution will be the final state for the tunneling calculation. Note that the point P indicates a closed anti-trapped surface, so this configuration could not emerge classically without an initial singularity. (The figure is drawn for $\gamma = .75$ and $M/M_{cr} = .3$.)

initial singularity would be avoided, and it becomes possible in principle to create a universe in the lab.

III. THE CLASSICAL ACTION

In order to quantize a system it is extremely useful to construct its classical action. For canonical quantization this action, with its associated Lagrangian, allows us to define canonical momenta and determine the Hamiltonian. In a path integral approach, the action appears directly in the expression for amplitudes.

The generic expression for the action describing matter fields interacting with gravity is the sum of a gravity term, a matter term, and a surface term:

$$I = \frac{1}{16\pi G} \int d^4x \sqrt{g} \mathcal{R} + \int d^4x \sqrt{g} \mathcal{L}_{\text{matter}} + \frac{1}{8\pi G} \int d^3x \sqrt{h} K. \quad (3.1)$$

We use conventions for which $g_{\mu\nu}$ has one negative eigenvalue, $g \equiv -\det g_{\mu\nu}$, $\mathcal{R} \equiv g^{\mu\nu} \mathcal{R}_{\mu\nu}$ is the Ricci scalar,²⁰ K is the extrinsic curvature of the three-dimensional boundary of the four-dimensional region of integration, and h is the determinant of the three-dimensional metric of the boundary.

If the surface term were omitted, the gravitational field equations could be obtained by extremizing the action with respect to variations for which the metric and its normal derivatives are held fixed on the boundary. With the surface term,²¹ the field equations can be obtained from a variational principle for which only the metric on the boundary is held fixed. This will be seen clearly in the specific problem to be analyzed.

In the thin-wall bubble problem the degrees of freedom have been reduced to just one: the radius of the bubble wall, τ . In order to reduce the action I to a functional depending only on the wall trajectory, we must associate a spacetime volume with each trajectory segment. Note that the action must be defined for arbitrary trajectories, and not just for solutions of the equations of motion.

To begin we will consider only trajectories that lie in the right quadrant of the Kruskal diagram and the left quadrant of the de Sitter diagram. For trajectories of this type, we illustrate our choice for the associated spacetime volume in Fig. 6. In the Schwarzschild region the past and future boundaries of the volume are taken as hypersurfaces of constant T_S , corresponding to lines running radially outward from the origin in the U_S - V_S plane. A boundary is also chosen at some asymptotically large value of R , which we call R_{B1G} . Since the variational principle requires the metric on the boundary to be fixed, and since the Schwarzschild time interval can be determined from the boundary at R_{B1G} , it follows that the Schwarzschild time interval must be held fixed. Without loss of generality we can fix both the initial and final Schwarzschild times, t_S^i and t_S^f . In the de Sitter region we similarly choose the past and future boundaries of the volume to be hypersurfaces of constant T_D , corresponding to lines running radially outward from the origin in the U_D - V_D plane.

Although we are interested in trajectories that span a fixed interval of Schwarzschild time t_S , it is nonetheless convenient to express many of the results in terms of the coordinate-invariant function $r(\tau)$. The relation between τ and t_S is determined by Eq. (2.12), which

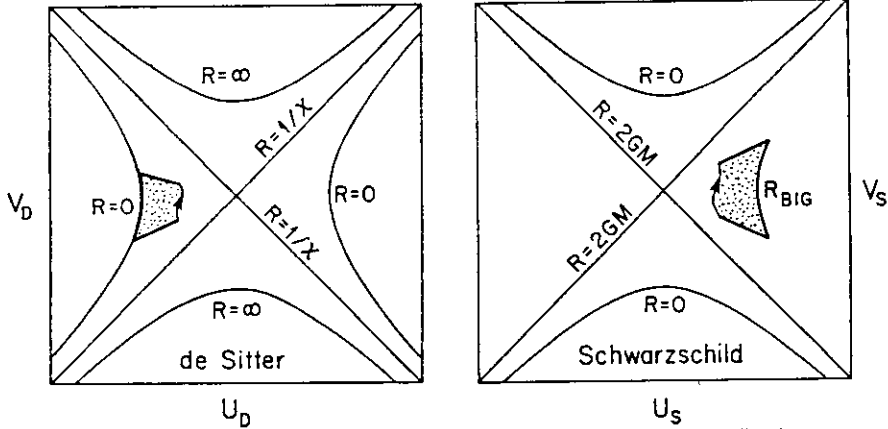


Fig. 6: The spacetime volume of integration associated with a segment of bubble wall trajectory. The trajectory under consideration lies in the right quadrant of the Kruskal diagram, and the left quadrant of the de Sitter diagram. The volume is shown as a dotted region, with a heavy line indicating its boundary.

can be rewritten as

$$\frac{dt_S}{d\tau} = \frac{\beta_S}{A_S}. \quad (3.2)$$

We will consider trajectories $r(\tau)$ from τ^i to τ^f , with the values of r at the end points fixed: i.e., $r(\tau^i) = r^i$ and $r(\tau^f) = r^f$. However, τ^f is not fixed, but is determined by

$$t_S^f - t_S^i = \int_{\tau^i}^{\tau^f} \frac{\beta_S}{A_S} d\tau, \quad (3.3)$$

with $t_S^f - t_S^i$ fixed. The de Sitter time interval is also determined by the trajectory, and again using Eq. (2.12) one finds

$$t_D^f - t_D^i = - \int_{\tau^i}^{\tau^f} \frac{\beta_D}{A_D} d\tau. \quad (3.4)$$

Clearly t_S^i , t_D^i , and τ^i are arbitrary, and could be set to zero.

To evaluate the action it is convenient to introduce Gaussian normal coordinates near the wall, for an arbitrary wall trajectory $r(\tau)$. First, specify points on the spacetime hypersurface of the wall by τ , θ , and φ , where τ is proper time measured in the wall and θ and φ are the usual angular variables. At each point P of the wall hypersurface, construct a geodesic normal to the surface (using the full four geometry), and assign to each point of the geodesic the same values of τ , θ , and φ as the point P . Define a normal coordinate η as the proper distance away from the wall along the geodesic. We take η as positive in the Schwarzschild region and negative in the de Sitter region. The metric has the form

$$ds^2 = g_{\tau\tau}(\tau, \eta) d\tau^2 + d\eta^2 + R^2(\tau, \eta) d\Omega^2, \quad (3.5)$$

where $g_{\tau\tau}(\tau, 0) = -1$ and $R(\tau, 0) = r(\tau)$. Here $R(\tau, \eta)$ plays the role of a function used to describe the metric; however, since it is the radius of the (θ, ϕ) two-sphere, it must agree with the coordinate R of the Schwarzschild or static de Sitter coordinate systems. The transformation to Gaussian normal coordinates can in fact be carried out in closed form, as we show in Appendix A.

We now evaluate the action (3.1), using the region of integration described above. We begin with the matter term, which has a contribution from the bubble wall, and also a contribution from the constant energy density ρ_f inside the bubble volume. In the thin-wall limit the internal structure of the bubble wall is neglected, and the wall action is proportional to the three-volume of the world hypersurface swept out by the motion of the wall. The constant of proportionality is $-\sigma$, where σ is the energy per unit area of the wall. Thus,

$$I_{\text{matter}}^{\text{wall}} = -4\pi\sigma \int_{\tau^i}^{\tau^f} r^2(\tau) d\tau. \quad (3.6)$$

In Appendix B we show that this expression gives the correct energy-momentum tensor for the wall. Using Eqs. (2.6) and (2.12), the bubble volume (BV) term can be written as

$$\begin{aligned} I_{\text{matter}}^{\text{BV}} &= \rho_f \int_{t_D^i}^{t_D^f} dt_D \int_0^{r(t_D)} dR \int d\theta d\varphi R^2 \sin\theta \\ &= \frac{4\pi\rho_f}{3} \int_{t_D^i}^{t_D^f} dt_D r^3(t_D) \\ &= -\frac{4\pi\rho_f}{3} \int_{\tau^i}^{\tau^f} r^3(\tau) \frac{\beta_D}{A_D} d\tau, \end{aligned} \quad (3.7)$$

where the change of sign in the last line is caused by the minus sign in Eq. (2.12).

We now turn to the gravitational term of the action. In the de Sitter region, the curvature scalar $\mathcal{R} = 12\chi^2 = 32\pi G\rho_f$, so the bubble volume contribution to the action is

$$\begin{aligned} I_{\text{gravity}}^{\text{BV}} &= -\frac{32\pi G\rho_f}{16\pi G} \int_{t_D^i}^{t_D^f} dt_D \int_0^{r(t_D)} dR \int d\theta d\varphi R^2 \sin\theta \\ &= \frac{8\pi\rho_f}{3} \int_{\tau^i}^{\tau^f} r^3(\tau) \frac{\beta_D}{A_D} d\tau. \end{aligned} \quad (3.8)$$

Note that (3.7) and (3.8) can be added to give

$$I^{\text{BV}} = \frac{4\pi\rho_f}{3} \int_{\tau^i}^{\tau^f} r^3(\tau) \frac{\beta_D}{A_D} d\tau. \quad (3.9)$$

In the Schwarzschild region $\mathcal{R} = 0$, so there is no contribution to the action.

There is also a contribution to the gravitational action from the bubble wall, where the geometry changes from de Sitter to Schwarzschild. It is most convenient to evaluate this in Gaussian normal coordinates:

$$I_{\text{gravity}}^{\text{wall}} = \frac{1}{16\pi G} \int_{-\epsilon}^{\epsilon} d\eta \int_{r_i}^{r_f} d\tau \int d\theta d\varphi \sqrt{g} \mathcal{R} \quad (3.10)$$

where ϵ is an arbitrarily small positive number. In Gaussian normal coordinates, the four-dimensional Ricci scalar can be expressed in terms of three-dimensional tensors (defined on hypersurfaces of constant η) and their normal derivatives:

$$\mathcal{R} = {}^3\mathcal{R} - (K_{ij}K^{ij} + K^2) - 2\frac{\partial}{\partial\eta}K \quad (3.11)$$

where $i, j = \tau, \theta, \varphi$, K_{ij} is the extrinsic curvature, $K \equiv K^i_i$, and ${}^3\mathcal{R}$ is the Ricci scalar constructed from the three-dimensional metric g_{ij} in the hypersurface of constant η . The extrinsic curvature is related to the normal derivative of the metric by

$$K_{ij} = \frac{1}{2} \frac{\partial g_{ij}}{\partial \eta}. \quad (3.12)$$

The value of K_{ij} for this case is calculated in Appendix A, and has also been calculated in Ref. [6]. It is given by

$$K_{\theta\theta} = \frac{K_{\varphi\varphi}}{\sin^2\theta} = \begin{cases} r\beta_D & \text{for } \eta = -\epsilon \\ r\beta_S & \text{for } \eta = +\epsilon \end{cases} \quad (3.13)$$

$$K_{\tau\tau} = \begin{cases} -\dot{\beta}_D/\dot{r} = -\frac{1}{\beta_D} \left(\dot{r} + \frac{1}{2} \frac{dA_D}{dr} \right) & \text{for } \eta = -\epsilon \\ -\dot{\beta}_S/\dot{r} = -\frac{1}{\beta_S} \left(\dot{r} + \frac{1}{2} \frac{dA_S}{dr} \right) & \text{for } \eta = +\epsilon \end{cases}.$$

Since the extrinsic curvature is discontinuous in η , the last term in Eq. (3.11) contributes a δ -function to $\mathcal{R}$. This is the only term which survives the integration over the thickness of the wall, giving

$$I_{\text{gravity}}^{\text{wall}} = \frac{1}{8\pi G} \int_{r_i}^{r_f} d\tau \int d\theta d\varphi r^2 \sin\theta K \Big|_{-\epsilon}^{+\epsilon}. \quad (3.14)$$

Using Eqs. (3.5) and (3.13) to calculate the discontinuity in K , one finds

$$I_{\text{gravity}}^{\text{wall}} = \int_{r_i}^{r_f} d\tau \left\{ \frac{1}{2G} \left[2r\beta + \frac{r^2}{\beta} \left(\dot{r} + \frac{1}{2} \frac{dA}{dr} \right) \right] \right\} \Big|_S^D. \quad (3.15)$$

The action (3.1), not including the surface term, is the sum of Eqs. (3.6), (3.9) and (3.15), or

$$I_{\text{gravity}} + I_{\text{matter}} = \int_{r_i}^{r_f} L_0 d\tau, \quad (3.16)$$

where

$$L_0 = -4\pi\sigma r^2 + \frac{1}{2G} \left[2r\beta + \frac{r^2}{\beta} \left(\dot{r} + \frac{1}{2} \frac{dA}{dr} \right) \right] \Big|_S^D + \frac{4\pi\rho_f}{3} r^3 \frac{\beta_D}{A_D}. \quad (3.17)$$

This action, when properly varied, yields the equation of motion for r . Recall that r^f is not held fixed, but is determined by (3.3). Since the action contains $\dot{r}$, it is necessary to keep both r and $\dot{r}$ fixed on the initial and final surfaces. However, since $\dot{r}$ appears linearly, the resulting equation of motion is a second order differential equation. The resulting equation is in fact the proper time derivative of Eq. (2.11).

Because of the $\dot{r}$ in the Lagrangian (3.17), it is difficult to quantize this system as it stands. From the canonical point of view it is not clear what the conjugate momentum is. From the path integral point of view, Gibbons and Hawking²¹ have pointed out that a Lagrangian containing second time derivatives leads to evolution amplitudes $\langle x_2 | U(t_2, t_1) | x_1 \rangle$ that violate the usual property of intermediate state insertion:

$$\langle x_2 | U(t_2, t_1) | x_1 \rangle = \int dx_3 \langle x_2 | U(t_2, t_3) | x_3 \rangle \langle x_3 | U(t_3, t_1) | x_1 \rangle, \quad (3.18)$$

where $t_1 < t_3 < t_2$. The property above is normally demonstrated in the path integral formulation by noting that any path contributing to $\langle x_3 | U(t_3, t_1) | x_1 \rangle$ can be combined with any path contributing to $\langle x_2 | U(t_2, t_3) | x_3 \rangle$ to obtain a path that contributes to $\langle x_2 | U(t_2, t_1) | x_1 \rangle$. One then uses the additivity of the action.

$$I(t_2, t_1) = I(t_2, t_3) + I(t_3, t_1), \quad (3.19)$$

to show that the contribution to $\langle x_2 | U(t_2, t_1) | x_1 \rangle$ is the product of the contribution to $\langle x_3 | U(t_3, t_1) | x_1 \rangle$ and the contribution to $\langle x_2 | U(t_2, t_3) | x_3 \rangle$; the result (3.18) then follows by summing over all paths. When the Lagrangian depends on $\dot{r}$, however, Eq. (3.19) fails. To see this, recall that the paths of interest are constructed by joining two arbitrary paths at $t = t_3$, and typically such combined paths have a discontinuity in $\dot{x}$ at $t = t_3$. Such a discontinuity leads to $\ddot{x} \propto \delta(t - t_3)$, which yields a contribution to $I(t_2, t_1)$, but no contribution to either $I(t_2, t_3)$ or $I(t_3, t_1)$. The surface term of Eq. (3.1) is constructed to remedy this discrepancy by producing contributions to $I(t_2, t_3)$ and $I(t_3, t_1)$ which sum to match the δ -function contribution to $I(t_2, t_1)$.

(We show in Appendix C a method by which any Lagrangian $L(x, \dot{x}, \ddot{x})$ that depends linearly on $\ddot{x}$ can be converted to a Lagrangian depending only on x and $\dot{x}$ by the addition of a boundary term to the action. When this recipe is applied to the problem at hand, it produces essentially the same modification of the Lagrangian as results from the surface

term of Eq. (3.1). The two recipes differ only by a dynamically irrelevant contribution arising from the surface at spatial infinity.)

The surface term can be written as

$$I_{\text{surface}} = \frac{1}{8\pi G} \int d^3x \sqrt{h} \nabla_\mu n^\mu, \quad (3.20)$$

where x^i are the coordinates of the three surface, h is the determinant of the metric in the three surface, n^μ is the unit normal to the three surface, and ∇_μ is the (four-dimensional) covariant derivative.²² Along the $T_S = \text{constant}$ part of the boundary, but away from the wall itself, the extrinsic curvature vanishes. To see this explicitly we can use the Schwarzschild coordinates $(T_S, R, \theta, \varphi)$, with the metric given by (2.1). Then $n_S^\mu = (A_S^{-1/2}, 0, 0, 0)$ and

$$\nabla_\mu n^\mu = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} n^\mu) = \frac{1}{\sqrt{g}} \frac{\partial}{\partial T_S} (\sqrt{g} n^{T_S}) = 0, \quad (3.21)$$

since g and A_S are independent of T_S . Similarly, there is no contribution to the surface term from the $T_D = \text{constant}$ part of the boundary.

At the wall, however, where the $T_S = \text{constant}$ surface and the $T_D = \text{constant}$ surface meet, the normal vector n^μ changes discontinuously. Care must be taken in evaluating the surface term (3.20) at this place, since $\nabla_\mu n^\mu$ contains a δ -function contribution concentrated at a point where the metric determinant h is discontinuous. To illustrate how to handle this problem we consider a simple analogous problem in two-dimensional flat Euclidean space. The “surface” we wish to consider consists of two straight line segments meeting at a point. The normal is discontinuous at the intersection yet we wish to evaluate

$$\int dx \sqrt{h} \nabla_\mu n^\mu. \quad (3.22)$$

The geometry is illustrated in Fig. 7a.

We first smooth out the sharp bend in some arbitrary way as is shown in Fig. 7b. Let $y = f(x)$ describe the smoothed out curve where x and y are the Cartesian coordinates of the flat two-dimensional space. The normal to the curve is²³

$$n^\mu = \frac{1}{\sqrt{1 + (df/dx)^2}} \left(-\frac{df}{dx}, 1 \right), \quad (3.23)$$

and the line element is

$$dx \sqrt{h} = dx \sqrt{1 + (df/dx)^2}. \quad (3.24)$$

The integration can be carried out exactly, giving

$$\int_{x_1}^{x_2} dx \sqrt{h} \nabla_\mu n^\mu = \theta_1 - \theta_2, \quad (3.25)$$

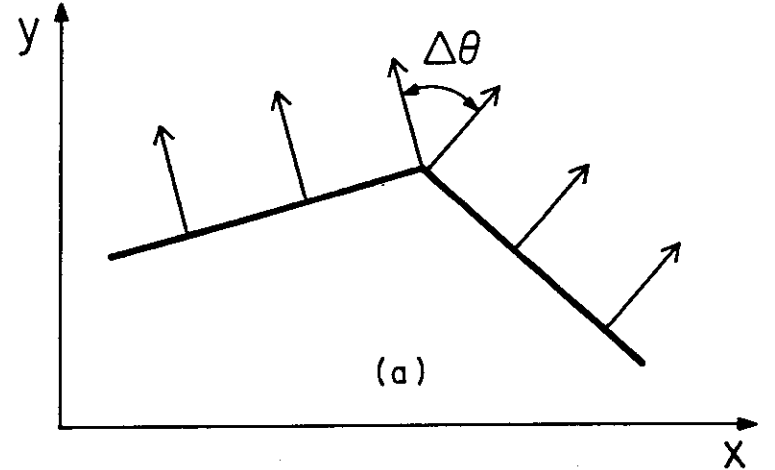


Fig. 7a: The sharp edge. To evaluate the surface term in the gravitational action for a surface with a discontinuous normal, we begin by evaluating an analogous expression for the two-dimensional flat space shown in this diagram.

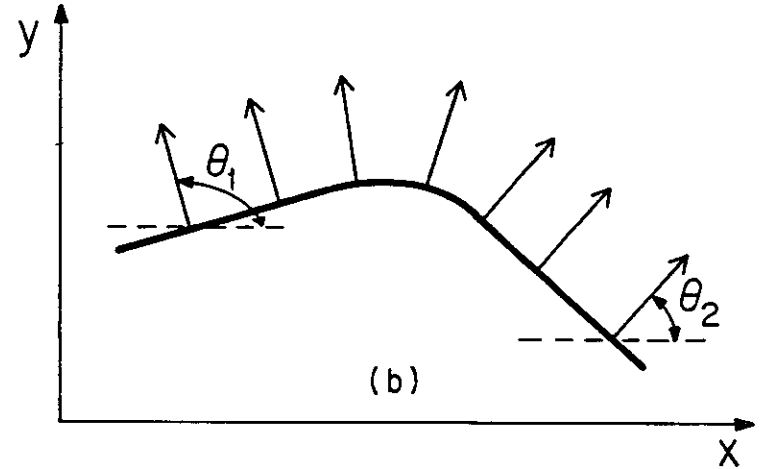


Fig. 7b: The edge smoothed out. To make the integral well defined, the line with the discontinuous normal is replaced by one with a continuous normal. The integration can be carried out by using an integration variable θ , defined (as shown) as the angle between the normal and the x -axis. At the end we take the limit in which the size of the smoothing region approaches zero.

where $\cot \theta = df/dx$, so θ is the angle that n^μ makes with respect to the x -axis. Thus, the integral is simply the change in the angle of the normal between the points x_1 and x_2 . Since the answer is independent of the detailed shape, there is no difficulty in taking the limit as the shape approaches the sharp edge. The answer can be written as

$$\int_{\text{edge}} dx \sqrt{h} \nabla_\mu n^\mu = \cos^{-1}(n_1 \cdot n_2), \quad (3.26)$$

where n_1 and n_2 are the normals on either side of the edge.

The problem can be formulated similarly in a 1+1-dimensional Minkowski space, except for some sign changes. If the surface is spacelike, the answer is

$$\int_{\text{edge}} dx \sqrt{h} \nabla_\mu n^\mu = \cosh^{-1}(-n_1 \cdot n_2). \quad (3.27)$$

This corresponds physically to the rapidity (rapidity $\equiv \tanh^{-1} v$) of an observer at rest (i.e., moving normal to the spacelike surface) on one side of the discontinuity, as seen by an observer at rest on the other side.

The generalization from flat space to curved space is no problem—although the surface has a discontinuous normal, it is nonetheless embedded in a space which is locally flat. Since the smoothing takes place in a region that is arbitrarily small, the previous derivation can be repeated without change, using locally flat coordinates. The coordinate-invariant expressions (3.26) and (3.27) are then seen to be valid in curved space.

We can now apply this to our four-dimensional bubble problem. The normal to the boundary is discontinuous at the wall, and we call this normal n_S on the Schwarzschild side and n_D on the de Sitter side. Because of the spherical symmetry, the angular dependence is trivial and the problem looks much like the 1+1-dimensional Minkowski case. Thus the contribution to the surface term, (3.20), at the wall is

$$I_{\text{surface}}^{\text{wall}} = \frac{1}{8\pi G} [4\pi r^2(r)] \cosh^{-1}(-n_S \cdot n_D) \Big|_{r_i}^{r_f}. \quad (3.28)$$

Recalling that $\cosh^{-1}(-n_S \cdot n_D)$ is the relative rapidity of stationary observers in the Schwarzschild and de Sitter regions, we can evaluate this quantity by calculating the rapidity of each observer relative to the wall. The situation is illustrated schematically in Fig. 8. The figure shows that the relative rapidity $\Delta\theta \equiv \cosh^{-1}(-n_S \cdot n_D)$ is given by $\theta_S - \theta_D$, where θ_S (θ_D) is the rapidity of a stationary Schwarzschild (de Sitter) observer relative to the wall. Working in Schwarzschild coordinates $(T_S, R, \theta, \varphi)$, the normal vector n_S is given by

$$n_S^\mu = (A_S^{-1/2}, 0, 0, 0), \quad (3.29)$$

and the velocity four-vector of the bubble wall is

$$\begin{aligned} U_{\text{wall}}^\mu &= (\dot{t}_S, \dot{r}, 0, 0) \\ &= \left(\frac{\beta_S}{A_S}, \dot{r}, 0, 0 \right). \end{aligned} \quad (3.30)$$

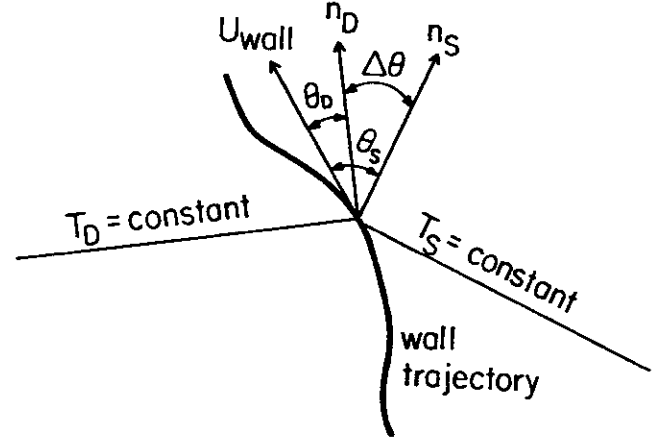


Fig. 8: Diagram for the calculation of the rapidity of an observer at rest in the Schwarzschild coordinate system, as viewed by an observer at rest in the de Sitter coordinate system. For illustrative purposes, normal vectors are shown as if the metric were Euclidean. Note that it suffices to calculate the rapidity of each observer relative to the wall.

Then

$$\theta_S = \cosh^{-1}(-n_S \cdot U_{\text{wall}}) = \cosh^{-1}(\beta_S / \sqrt{A_S}) = \tanh^{-1}(\dot{r} / \beta_S). \quad (3.31)$$

The calculation of θ_D proceeds identically, and finally one has

$$I_{\text{surface}}^{\text{wall}} = -\frac{1}{2G} \left\{ r^2 \tanh^{-1} \left(\frac{\dot{r}}{\beta} \right) \Big|_S^D \right\} \Big|_{r_i}^{r_f}. \quad (3.32)$$

In order to combine this term with the other terms in the action, it is useful to write it as the integral of its derivative:

$$I_{\text{surface}}^{\text{wall}} = -\frac{1}{2G} \int_{r_i}^{r_f} \frac{d}{dr} \left\{ r^2 \tanh^{-1} \left(\frac{\dot{r}}{\beta} \right) \Big|_S^D \right\} dr. \quad (3.33)$$

There are two more contributions to the surface term. One comes from the $R = \text{constant}$ surface at R_{BIG} and the other from its intersection with the $T_S = t_S^i$ and $T_S = t_S^f$ surfaces, as can be seen from Fig. 6. We can define a unit normal, called m , to the constant R surfaces. In static Schwarzschild coordinates $(T_S, R, \theta, \varphi)$

$$m^\mu = (0, A_S^{1/2}, 0, 0), \quad (3.34)$$

and

$$I_{\text{surface}}^{R_{\text{BIG}}} = \frac{1}{8\pi G} \int_{t_S^i}^{t_S^f} dt_S \int d\theta d\varphi A_S^{1/2} R^2 \sin \theta \nabla_\mu m^\mu \Big|_{R_{\text{BIG}}} \quad (3.35)$$

$$= \left(\frac{R_{\text{BIG}}}{G} - \frac{3M}{2} \right) (t_S^f - t_S^i), \quad (3.36)$$

which blows up as R_{BIG} goes to infinity. This is usually handled by a subtraction which renders it finite. The Gibbons–Hawking prescription²¹ is to subtract

$$(I_{\text{surface}})_0 \equiv \int d^3x \sqrt{h} K_0, \quad (3.37)$$

where K_0 represents the extrinsic curvature of the surface $R = R_{\text{BIG}}$ embedded in flat space. The result is

$$I_{\text{net}}^{R_{\text{BIG}}} = I_{\text{surface}}^{R_{\text{BIG}}} - (I_{\text{surface}}^{R_{\text{BIG}}})_0 = -\frac{M}{2} (t_S^f - t_S^i) + \mathcal{O}\left(\frac{1}{R_{\text{BIG}}}\right). \quad (3.38)$$

The edge contribution is canceled exactly by this prescription. In any case, with or without the subtraction, these terms are dynamically irrelevant. Their contribution to the action is a constant times $(t_S^f - t_S^i)$, and since we only make variations that keep the Schwarzschild time fixed, they do not affect the classical motion. The Hamiltonian (defined with respect to t_S) is modified only by the addition of a constant, so these terms are irrelevant for the quantum problem as well. Nonetheless we will keep the subtracted result (3.38), in order to conform to standard conventions. The full action is obtained by combining the expressions in (3.16), (3.33), and (3.38). To combine the terms compactly, it is useful to rewrite the time interval in (3.38) by using (3.3), and we also use $\kappa \equiv 4\pi G\sigma$. The full action can then be written as

$$I = \int_{r^i}^{r^f} L dr, \quad (3.39)$$

where after some algebra we find

$$L = \frac{r}{G} [\beta_D - \beta_S - \kappa r] - \frac{r\dot{r}}{G} \left[\tanh^{-1} \left(\frac{\dot{r}}{\beta} \right) \right] \Big|_S^D - M \frac{\beta_S}{A_S}. \quad (3.40)$$

The last term in Eq. (3.40) contributes a constant times $t_S^f - t_S^i$ to the action, and is therefore dynamically irrelevant.

Thus, the surface term did its job of eliminating $\bar{r}$ from the Lagrangian. To obtain the classical equations of motion from the variational principle, it is no longer necessary to keep $\dot{r}$ fixed at the endpoints. The evaluation of the equations of motion from the action (3.39) is carried out in Appendix D.

As a check we can take the $G \rightarrow 0$ limit of Eq. (3.40), finding

$$L_{G \rightarrow 0} = -4\pi\sigma r^2 - \frac{4\pi}{3} \rho_f r^3 \sqrt{\dot{r}^2 + 1}. \quad (3.41)$$

This expression agrees exactly with the $G \rightarrow 0$ limit of the $\mathcal{L}_{\text{matter}}$ term of the action (3.1), as one might hope. Note, however, that the Einstein–Hilbert action ($\sqrt{g}\mathcal{R}$) does not vanish in the $G \rightarrow 0$ limit, but contains nonvanishing terms that are canceled by the surface term. If we had not included $I_{\text{net}}^{R_{\text{BIG}}}$ in the action, then Eq. (3.41) would have been modified by

the appearance of the ugly but dynamically irrelevant term, $\frac{1}{2}M\sqrt{1 + \dot{r}^2}$, on the right-hand side.

Since we are keeping the Schwarzschild time interval fixed it is useful to rewrite the action (3.39) explicitly in terms of the Schwarzschild time variable. Defining

$$r' \equiv \frac{dr}{dt_S}, \quad (3.42)$$

we write

$$I = \int_{t_S^i}^{t_S^f} L_S(r, r') dt_S, \quad (3.43)$$

where

$$L_S(r, r') = L(r, \dot{r}) \frac{A_S}{\beta_S}, \quad (3.44)$$

since $dr/dt_S = A_S/\beta_S$. On the right-hand side of (3.44), $\dot{r}$ is expressed in terms of r and r' through the relation:

$$\dot{r} = \frac{A_S^{1/2} r'}{\sqrt{A_S^2 - r'^2}}. \quad (3.45)$$

The final expression can be expressed in terms of r and r' by using the above equation with (2.14), giving

$$\beta_S = \frac{A_S^{3/2}}{\sqrt{A_S^2 - r'^2}} \quad (3.46a)$$

$$\beta_D = \frac{\beta_S}{A_S} \sqrt{A_D A_S + r'^2 (1 - A_D/A_S)}. \quad (3.46b)$$

(Note that if we were interested in trajectories that lie outside the right-hand Kruskal quadrant and the left-hand de Sitter quadrant, then the signs of Eqs. (3.45) and (3.46) would be altered, as discussed immediately after Eq. (2.14).) In terms of these quantities L_S can be written as

$$L_S = \frac{r A_S}{G \beta_S} [\beta_D - \beta_S - \kappa r] + \frac{r r'}{G} \left[\tanh^{-1}(r'/A_S) - \tanh^{-1} \left(\frac{r' \beta_S}{A_S \beta_D} \right) \right] - M. \quad (3.47)$$

The action (3.47) can be varied, keeping r fixed at the initial and final Schwarzschild times (which are also held fixed), to give the second order equation of motion for r as in any ordinary mechanics problem. This equation is the same as the one obtained by varying (3.16), and is equivalent to the proper time derivative of (2.11). The conserved Hamiltonian H corresponding to this Lagrangian is displayed in Appendix D. The equation $H = M$ is

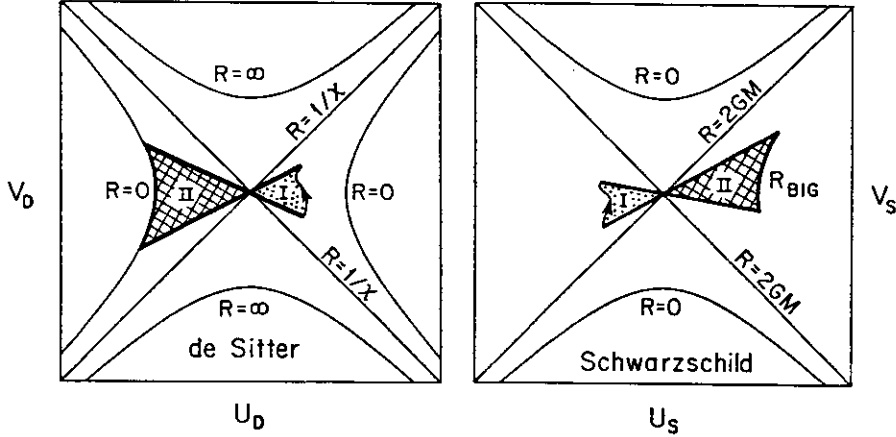


Fig. 9: The spacetime volume of integration associated with a segment of bubble wall trajectory in the left quadrant of the Kruskal diagram, and/or the right quadrant of the de Sitter diagram. The volume shown as a dotted region (region *I*) is to be counted positively, while the volume shown cross-hatched (region *II*) is to be counted negatively. The boundary is shown as a heavy line.

equivalent to the equation of motion (2.11), but for other values of the conserved Hamiltonian the solutions are unphysical, and do not represent solutions to the full field equations. The reason they arise here is that by reducing the degrees of freedom to just one, r , we have lost equations of motion. If, for example, we had kept g_{00} as a degree of freedom, then varying with respect to g_{00} would have given a constraint equation equivalent to (2.11).

So far we have succeeded in associating a spacetime volume with, and have calculated the action for, an arbitrary trajectory that lies in the right quadrant of the Kruskal diagram and the left quadrant of the de Sitter diagram. We now wish to allow for the possibility that the trajectory might lie in the left quadrant of the Kruskal diagram and/or the right quadrant of the de Sitter diagram, so for example the type (b) solution shown in Fig. 5 can be included. Our choice of a spacetime volume for these cases is illustrated in Fig. 9. Note that the $T_S = t_S^i$ surface and the $T_S = t_S^f$ surface cross each other, as do the $T_D = t_D^i$ and the $T_D = t_D^f$ surfaces. If we use the convention that the action of the wall itself and the nearby volumes (regions *I* in Fig. 9) is positive, then we will define the action on the far sides of the crosses (regions *II*) to be negative. By this we mean that we calculate the actions in regions *I* and *II* by using Eq. (3.1), but we multiply the expressions for regions *II* by -1 before adding them in to get the total action. This rule leads to an action which gives the correct equations of motion and is the proper way to handle a boundary which crosses itself. We will justify the rule after we first apply it to calculate the action.

To calculate the action for the space-time region shown in Fig. 9, we first adopt the convention that the proper time τ is increasing along the trajectory, so $\tau^f > \tau^i$. The action at the wall is just as in the case previously treated, so we have

$$I^{\text{wall}} = I^{\text{wall}}_{\text{matter}} + I^{\text{wall}}_{\text{gravity}} + I^{\text{wall}}_{\text{surface}}, \quad (3.48)$$

where the three terms on the right-hand side come from (3.6), (3.15) and (3.33). Similarly, the contribution from the surface term at $R = R_{\text{BIG}}$ is the same as before, so the action includes the expression for $I^{\text{BIG}}_{\text{net}}$ from Eq. (3.38). For the bubble volume, recall that for a trajectory in the left quadrant the matter and gravity contributions are proportional, and add to give (3.9). For a trajectory in the right quadrant, the bubble volume consists of both regions *I* and *II* on the de Sitter side of Fig. 9. The region *I* bubble volume contribution is then

$$I^{\text{BV}}_I = -\frac{4\pi\rho_f}{3} \int_{\tau^i}^{\tau^f} \left[\frac{1}{\chi^3} - r^3(\tau) \right] \frac{\beta_D}{A_D} d\tau. \quad (3.49)$$

Since $\beta_D < 0$ and $r < \chi^{-1}$, the above expression is positive, as is the right-hand side of Eq. (3.9) when applied to a trajectory in the left quadrant. The region *II* bubble volume contribution, including the -1 from our rule, is

$$I^{\text{BV}}_{II} = \frac{4\pi\rho_f}{3} \int_{\tau^i}^{\tau^f} \frac{1}{\chi^3} \frac{\beta_D}{A_D} d\tau, \quad (3.50)$$

which is negative. The total bubble volume action is then

$$I^{\text{BV}} = \frac{4\pi\rho_f}{3} \int_{\tau^i}^{\tau^f} r^3(\tau) \frac{\beta_D}{A_D} d\tau, \quad (3.51)$$

which has the same form as (3.9). The total action is the sum of Eqs. (3.48), (3.38), and (3.51), which gives an expression for L identical to the previous case, Eqs. (3.39) and (3.40). This guarantees that this action gives the correct equations of motion.

To understand the rule for the sign of the action when the boundary of the region crosses itself, we first note that the same rule is used in discussing an ordinary definite integral. When one says that

$$\int_a^b f(x) dx$$

represents the area between the curve $f(x)$ and the x -axis, one is using the convention that the area is counted negatively if the curve $f(x)$ crosses the x -axis and becomes negative.

To understand the rule in a physical context, we consider a simple two-dimensional Euclidean scalar field theory example, with an action that depends on second derivatives:

$$I_V = \frac{1}{2} \int_M d^2x \phi \nabla^2 \phi. \quad (3.52)$$

M is some region in $\mathbb{R}^2$, as shown in Fig. 10a. By varying (3.52) we get the equation of motion $\nabla^2 \phi = 0$, but the variation $\delta\phi$ must vanish at the boundary of M , ∂M , as must the normal derivative $\hat{n} \cdot \nabla \delta\phi$, where $\hat{n}$ is an outward normal vector on the boundary.

We can remove the restriction that the normal derivative of the variation vanish by adding a surface term:

$$I_{\text{surface}} = -\frac{1}{2} \int_{\partial M} \phi \nabla \phi \cdot d\hat{n} \quad (3.53a)$$

$$= -\frac{1}{2} \int_M d^2x \nabla \cdot (\phi \nabla \phi). \quad (3.53b)$$

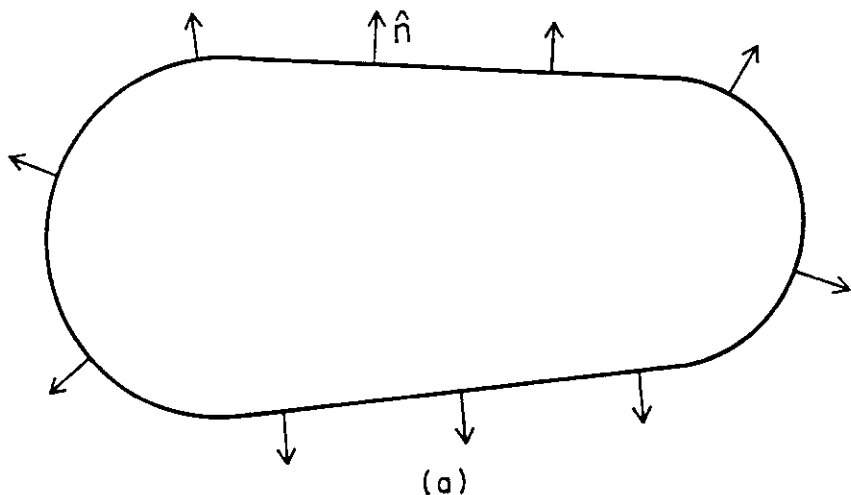


Fig. 10a: A simple region of integration M , with the unit normal vector $\hat{n}$ pointing outward.

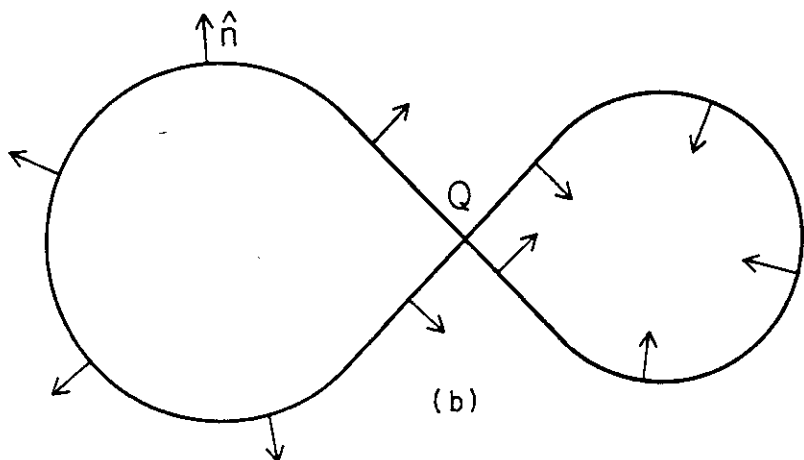


Fig. 10b: A region of integration M with a boundary that crosses itself at the point Q . If the normal vector $\hat{n}$ is to be continuous, then it must switch at Q from pointing outward to pointing inward.

Then

$$I = I_V + I_{\text{surface}} = -\frac{1}{2} \int_M d^2x (\vec{\nabla} \phi)^2 \quad (3.54)$$

will yield the correct equation of motion if ϕ is varied with $\delta\phi$ fixed on ∂M , with no restriction on $\hat{n} \cdot \vec{\nabla} \phi$.

Now consider a case in which the boundary crosses itself at some point Q , as shown in Fig. 10b. The region M is then divided into two regions, labeled I and II in the figure. If the normal vector $\hat{n}$ is required to vary continuously along the boundary, then it switches at Q from pointing outward to pointing inward. If Eqs. (3.53a) and (3.53b) are to be equivalent for this case, then the volume integral of (3.53a) must be defined so that the contribution from region II is counted negatively. Thus, it is necessary to adopt these sign conventions for the action in order to obtain the correct equations of motion.

We have derived an expression for an action which is a functional of wall trajectories which lie in either the left or right quadrants of the Kruskal diagram and at the same time lie in the left or right quadrants of the de Sitter diagram. Our derivation, which associates a spacetime volume with a trajectory, breaks down if the trajectory enters either the upper or lower quadrant of either the Kruskal or de Sitter diagrams. In fact the expression (3.40) is not defined if this happens, because in the upper and lower quadrants $|\dot{r}/\beta| > 1$. We can write down an expression which gives the correct equations of motion in this case by replacing the inverse hyperbolic tangent by the real part of its analytic continuation, which is $\tanh^{-1}(\beta/\dot{r})$ when $|\dot{r}/\beta| > 1$. (The analytic continuation also gives an imaginary part, but its contribution to the Lagrangian is a total derivative which does not affect the equations of motion.) Although this action gives the correct equations of motion we do not have a geometric interpretation for it.

IV. THE SUBTRACTED TUNNELING ACTION

We would like to quantize the thin-wall bubble whose classical motion we have described above. Of course this problem involves gravity, and no tractable theory of quantum gravity exists. However, since we have reduced the problem to a single degree of freedom, r , that is governed by a known Lagrangian (3.47), the natural place to begin is the quantization of this system. Fortunately, we will not be interested in all aspects of this problem; we only wish to calculate the first term in the WKB expansion of a particular tunneling rate. Although the quantization will be ambiguous due to uncertainties in operator orderings, we will argue that these ambiguities do not affect the leading WKB approximation.

In this section we present a formal derivation of the tunneling amplitude, in leading WKB approximation. The answer will be of the form $\exp(-I_E/\hbar)$, where I_E is what we call a subtracted action. It is the action of a Euclidean solution that evolves from the initial state of the tunneling to the final state, minus the action of a Euclidean configuration (not a solution) that remains static at the initial state of the tunneling, enduring for the same interval of Euclidean time as the solution. In Secs. V and VI we apply this general formalism to the thin-wall false vacuum bubble tunneling problem, using a functional integral approach (Sec. V) and a purely canonical approach (Sec. VI). We find that each of these approaches

leads to ambiguities that we have not fully resolved, but the functional integral approach yields an answer which we think is correct.

The starting point for the formal derivation is the classical Lagrangian, L_S , which is a function of r and $r' \equiv dr/dt_S$. We then define as usual the momentum,

$$p \equiv \frac{\partial L_S}{\partial r'} , \quad (4.1)$$

and the Hamiltonian,

$$H \equiv pr' - L_S(r, r') . \quad (4.2)$$

H is a constant of the motion as usual, but it cannot take on arbitrary values. Because we are working with only one degree of freedom, r , we have lost equations of motion. In particular there is a constraint equation, equivalent to (2.11), which fixes H to have a particular value which we will call E :

$$H(p, r) = E . \quad (4.3)$$

(For the case of interest $E = M$, but the value of E is immaterial to the present discussion.)

To quantize this system we promote p , r , and H to operators and write the Schrödinger equation

$$i\hbar \frac{\partial}{\partial t_S} \Psi(r, t_S) = \hat{H} \Psi(r, t_S) , \quad (4.4)$$

where $\hat{H} \equiv H(\hat{p}, \hat{r})$, with $\hat{p} \equiv -i\hbar \frac{\partial}{\partial r}$. We also have the constraint equation

$$\hat{H} \Psi(r, t_S) = E \Psi(r, t_S) , \quad (4.5)$$

which means that Ψ has a trivial time dependence

$$\Psi(r, t_S) = e^{-iEt_S/\hbar} \Psi_E(r) , \quad (4.6)$$

where

$$\hat{H} \Psi_E(r) = E \Psi_E(r) , \quad (4.7)$$

and again E can have only one value. (Eq. (4.7) is a special case of the Wheeler-DeWitt equation.)

The definition of the quantum Hamiltonian, $\hat{H} \equiv H(\hat{p}, \hat{r})$, is frequently plagued by ordering ambiguities, but to lowest order in the WKB expansion these ambiguities can be ignored. To see this, imagine that $H(p, r)$ can be expanded in a Taylor series:

$$H(p, r) = \sum_{n=0}^{\infty} a_n(r) p^n . \quad (4.8)$$

We use this to define the quantum Hamiltonian, allowing for an arbitrary resolution of the ordering ambiguities:

$$\hat{H} = \sum_{n=0}^{\infty} \sum_m c_m \mathcal{O}_{n,m} , \quad (4.9)$$

where $\mathcal{O}_{n,m}$ represents some ordering of the operator product $a_n(\hat{r}) \hat{p}^n$, and the arbitrary coefficients c_m are restricted by the condition $\sum_m c_m = 1$. If we define $\sigma(r)$ by

$$\psi_E(r) \equiv e^{i\sigma(r)/\hbar} \quad (4.10)$$

and substitute into (4.7), then to lowest order in $\hbar$:

$$\sum_{n=0}^{\infty} a_n(r) \left(\frac{d\sigma}{dr} \right)^n = E , \quad (4.11)$$

or

$$H \left(\frac{d\sigma}{dr}, r \right) = E . \quad (4.12)$$

This is the classical Hamilton-Jacobi equation. The solution $\sigma(r)$, known as Hamilton's characteristic function, is given by

$$\sigma(\bar{r}) = \int^{\bar{r}} p(r) dr , \quad (4.13)$$

where $p(r)$ satisfies the classical equation (4.3), and the argument of σ has been indicated with an overbar to distinguish it from the variable of integration on the right-hand side.

The classically allowed region consists of those values of r for which (4.3) can be solved with real values of p ; other values of r are called forbidden. A turning point, r_1 , is the boundary between an allowed and forbidden region. Since a classical solution exists in the allowed region but cannot enter the forbidden region, we have $dr/dt_S = 0$ at r_1 . Using Eqs. (4.2) and (4.3), we find

$$H = -L_S(r_1, 0) = E . \quad (4.14)$$

In the allowed region one can find a solution $r(t_S)$ to the classical equation of motion, and then Eq. (4.13) can be rewritten as

$$\begin{aligned} \sigma(\bar{r}) &= \int^{\bar{r}} p(r) dr \\ &= \int^{\bar{r}} p \frac{dr}{dt_S} dt_S = \int^{\bar{r}} (L_S + H) dt_S \\ &= \int^{\bar{r}} \{ L_S(r, r') - L_S(r_1, 0) \} dt_S . \end{aligned} \quad (4.15)$$

In the classically forbidden region we assume that there exists a solution $r(t_{SE})$ to the classical equations of motion, where t_{SE} is the Euclidean time variable

$$t_{SE} \equiv it_S . \quad (4.16)$$

(For the false vacuum bubble problem, the existence of this solution is apparent from the equation of motion in the form (2.15).) Using this solution to rewrite Eq. (4.13) as above, one finds

$$\sigma(\bar{r}) = i \int^{\bar{r}} \left\{ L_{SE} \left(r, \frac{dr}{dt_{SE}} \right) - L_{SE}(r_1, 0) \right\} dt_{SE}, \quad (4.17)$$

where $r(\bar{t}_{SE}) = \bar{r}$, and L_{SE} is the Euclidean Lagrangian, defined by

$$L_{SE} \left(r, \frac{dr}{dt_{SE}} \right) \equiv -L_S \left(r, i \frac{dr}{dt_{SE}} \right). \quad (4.18)$$

(The minus sign in front of the right-hand side of the above definition is included so that a Lagrangian of the standard form, $L = \frac{1}{2}m\dot{x}^2 - V(x)$, analytically continues to give $L_E = \frac{1}{2}m(dx/dt_E)^2 + V(x)$.) If we enter a classically forbidden region at r_1 and exit at r_2 , then the ratio of amplitudes at r_1 and r_2 is

$$\frac{\psi(r_2)}{\psi(r_1)} \approx \exp \left[-\frac{I_E}{\hbar} \right], \quad (4.19)$$

where

$$I_E = \int_0^{\bar{t}_{SE}} \left\{ L_{SE} \left(r, \frac{dr}{dt_{SE}} \right) - L_{SE}(r_1, 0) \right\} dt_{SE}, \quad (4.20)$$

and $\bar{t}_{SE}$ is the Euclidean transit time through the forbidden region.

Thus, the tunneling exponent is found by computing the action of the Euclidean solution, and then subtracting the action of a Euclidean trajectory that remains static at the turning point, evolving for the same interval of Euclidean time as the solution. This method is very convenient, because it avoids the need to explicitly calculate either the canonical momentum or the Hamiltonian. Note that the subtraction in Eq. (4.20) is not surprising—it is necessary, for example, so that the answer will be unchanged if a constant is added to the Lagrangian.

Note also that the problem under study is time-reversal invariant, so for any classical solution $r(t_{SE})$ there exists another solution, $r(-t_{SE})$. The two solutions will give opposite signs for the action I_E , so Eq. (4.19) could indicate either exponential suppression or exponential growth in the forbidden region. This is expected, since the Schrödinger equation can describe tunneling across the barrier in either direction. Recognizing this sign ambiguity, we rewrite Eq. (4.19) as

$$\frac{\psi(r_2)}{\psi(r_1)} \approx \exp \left[\pm \frac{I_E}{\hbar} \right], \quad (4.21)$$

To describe tunneling from r_1 to r_2 , we choose the solution with exponential suppression.

V. THE FUNCTIONAL INTEGRAL APPROACH TO THE THIN-WALL BUBBLE

In this section we evaluate the tunneling amplitude of Eqs. (4.20) and (4.21), using a functional integral approach. We find, however, that the Euclidean solution folds back on

itself in a manner which implies that it is not a manifold. We nonetheless propose a recipe for defining the action of the Euclidean interpolation, using the (analytically continued) classical equations of motion as a guide. We will conjecture that the action defined in this way gives a valid WKB approximation for the tunneling, and we will proceed to calculate it.

The functional integral approach to quantum gravity is a generalization of the usual Feynman sum-over-histories approach to quantum mechanics. The object of interest is the amplitude to go from a given initial slice (three-dimensional hypersurface) to a given final slice. Each slice is specified by a three-metric, h_{ij} , along with a set of quantum fields which we denote generically by ϕ , all of which are defined as functions of the hypersurface coordinates. Using primes to denote the functions describing the final slice, the amplitude is defined to be

$$A(h_{ij}, \phi, h'_{ij}, \phi') \equiv \int [dg_{\mu\nu}] [d\Phi] e^{iI/\hbar}, \quad (5.1)$$

where the integral is over all four geometries $g_{\mu\nu}$ that are bounded by two slices with metrics h_{ij} and h'_{ij} , and over all fields Φ that take the values ϕ and ϕ' on the two boundaries.

There are all sorts of difficulties in defining the functional integration of Eq. (5.1), associated with the non-renormalizability of conventional gravity and the related problem of specifying the measure over all geometric histories. We will not address these issues. We hope, however, that Eq. (5.1) can nonetheless be used to extract a valid semiclassical approximation, in which the functional integral is assumed to be dominated by a single path satisfying classical equations of motion.

The dominance of the functional integral by a single path is the small $\hbar$ approximation, equivalent to the WKB approximation. We are working in the thin-wall limit, and we assume that the stationary points maintain the thin-wall form. Since this is a tunneling problem, there is no real-time solution to the classical equations that interpolates between the slices. Based on our experience with quantum mechanics, however, we believe that the amplitude can be approximated by using an imaginary-time, or Euclidean, interpolating solution. Thus we write

$$A(h_{ij}, \phi, h'_{ij}, \phi') \sim e^{-I_E/\hbar} \quad (5.2)$$

where I_E is the properly subtracted Euclidean action of the Euclidean history which interpolates between the initial and final slices. The general form for the subtraction is shown in Eq. (4.20).

Specifically, we wish to estimate a tunneling rate from a type (a) trajectory to a type (b) trajectory. The form of the trajectories depends on the mass of the bubble, and we begin by considering the case $M_S < M < M_{cr}$. A trajectory of type (a) is shown in the upper part of Fig. 11. We have used the time translation invariance of the solution to set $t_S = 0$ and $t_D = 0$ when r reaches its maximum value, r_1 . The initial slice for the tunneling problem is taken to be the $T_S = 0$ surface in the Schwarzschild region and the $T_D = 0$ surface in the de Sitter region, as shown in the diagram. Note that the extrinsic curvature of the initial slice vanishes everywhere—it vanishes at the wall as a consequence of (3.32), with $\dot{r} = 0$, and it vanishes elsewhere as a consequence of (3.21). A trajectory of type (b) is shown in the lower part of Fig. 11, and in this case we have set $t_S = 0$ and $t_D = 0$ when r reaches its minimum value, r_2 . The final slice is again taken along the $T_S = 0$ and

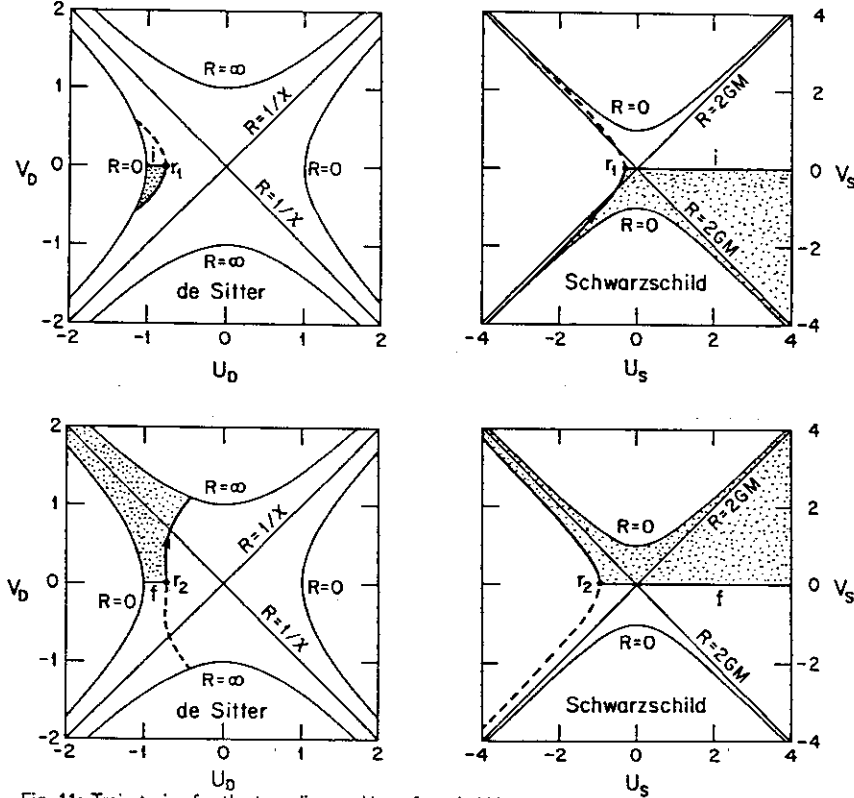


Fig. 11: Trajectories for the tunneling problem, for a bubble mass in the range $M_S < M < M_{cr}$. The upper diagrams show the initial trajectory, of type (a). Full de Sitter and Schwarzschild spaces are shown, with dotted regions indicating the physical spacetime. The initial slice for the tunneling problem is shown as a horizontal line labeled *i*. The lower diagrams indicate the final trajectory, of type (b), with the final slice shown as a horizontal line labeled *f*. (This figure and the following three are drawn for $\gamma = 1.95$ and $M/M_{cr} = .97$.)

$T_D = 0$ surfaces, so the extrinsic curvature again vanishes everywhere. The vanishing of the extrinsic curvature on the initial and final slices is the geometric generalization of the condition that $\dot{q} = 0$ at the turning points marking the beginning and end of a tunneling trajectory in a one-degree-of-freedom quantum mechanics problem.

We must now evaluate the action of Eq. (4.20) by constructing the Euclidean four-geometry that interpolates between the two three-geometries that we have described. First we need to describe Euclidean Schwarzschild and Euclidean de Sitter space.

Euclidean Schwarzschild space is described by the analytic continuation of (2.1):

$$ds_E^2 = A_S dT_{SE}^2 + A_S^{-1} dR^2 + R^2 d\Omega^2. \quad (5.3)$$

Here $R \geq 2GM$ and T_{SE} is a periodic variable with period $8\pi GM$. These coordinates cover all of Euclidean Schwarzschild space. Using Euclidean Kruskal-Szekeres coordinates, V_{SE}

and U_{SE} , the metric can be written as

$$ds_E^2 = 32 \frac{(GM)^3}{R} \exp\left(-\frac{R}{2GM}\right) (dV_{SE}^2 + dU_{SE}^2) + R^2 d\Omega^2, \quad (5.4)$$

where

$$\left(\frac{R}{2GM} - 1\right) \exp\left(\frac{R}{2GM}\right) = U_{SE}^2 + V_{SE}^2. \quad (5.5)$$

The Schwarzschild time variable is related to the angle in the U_{SE} - V_{SE} plane, with

$$T_{SE} = 4GM \tan^{-1}(V_{SE}/U_{SE}). \quad (5.6)$$

Note that $R = 2GM$ at the origin of the U_{SE} - V_{SE} plane, and grows monotonically as one moves outward. Further relations between these coordinates are given in Appendix E.

Euclidean de Sitter space is a four-sphere with radius $1/\chi$. The analytic continuation of (2.6) is

$$ds_E^2 = A_D T_{DE}^2 + A_D^{-1} dR^2 + R^2 d\Omega^2, \quad (5.7)$$

where $0 \leq R \leq 1/\chi$ and T_{DE} is a periodic variable with period $2\pi/\chi$. The continuation of the Gibbons-Hawking form of the metric (2.8) is

$$ds_E^2 = \chi^{-2} (1 + \chi R)^2 (dV_{DE}^2 + dU_{DE}^2) + R^2 d\Omega^2, \quad (5.8)$$

where

$$\frac{1 - \chi R}{1 + \chi R} = U_{DE}^2 + V_{DE}^2. \quad (5.9)$$

The static time variable is related to the angle in the U_{DE} - V_{DE} plane, with

$$T_{DE} = \chi^{-1} \tan^{-1}(V_{DE}/U_{DE}). \quad (5.10)$$

The entire space is represented by a disk $U_{DE}^2 + V_{DE}^2 \leq 1$ in the U_{DE} - V_{DE} plane. Each point in the disk represents a two-sphere of radius R , where R varies from 0 at the boundary to $1/\chi$ at the origin. More information about these coordinates is also given in Appendix E.

To construct the Euclidean history we need the Euclidean equation of motion for the bubble wall. It will be useful to introduce Euclidean proper time, τ_E , as well as the Euclidean Schwarzschild and de Sitter times, t_{SE} and t_{DE} , of the wall. Accordingly we define

$$\tau_E \equiv i\tau; \quad t_{SE} \equiv it_S; \quad t_{DE} \equiv it_D; \quad \text{and} \quad \dot{\tau}_E \equiv \frac{d\tau}{d\tau_E}, \quad (5.11)$$

as well as

$$\beta_{SE} = A_S \frac{dt_{SE}}{d\tau_E} = \pm \sqrt{A_S - \dot{\tau}_E^2} \quad (5.12a)$$

$$\beta_{DE} = -A_D \frac{dt_{DE}}{d\tau_E} = \pm \sqrt{A_D - \dot{\tau}_E^2}. \quad (5.12b)$$

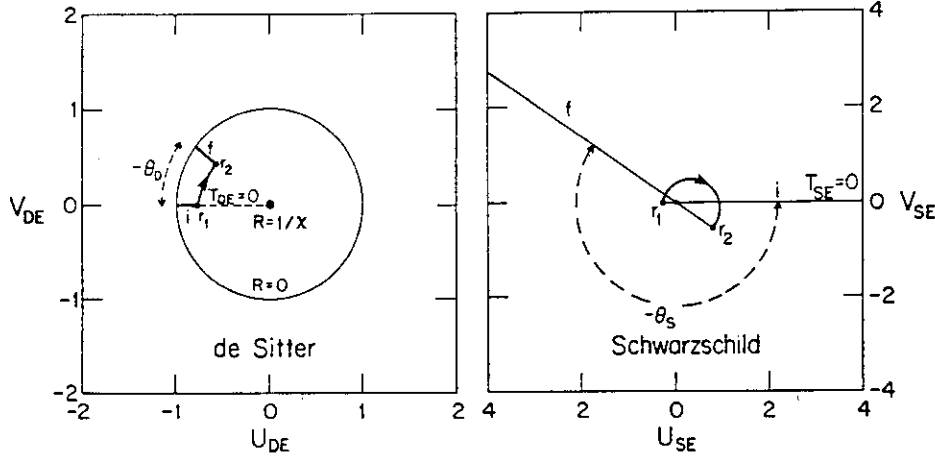


Fig. 12: The bubble wall trajectory in the Euclidean de Sitter and Schwarzschild spaces, for $M_S < M < M_{cr}$. The Euclidean de Sitter space is shown in Euclidean Gibbons-Hawking coordinates, and the Euclidean Schwarzschild space is shown in Euclidean Kruskal-Szekeres coordinates. The initial hypersurface is labeled i , the final hypersurface is labeled f , and the bubble wall trajectory is shown as a heavy line with arrows on it. The final hypersurface is oriented at an angle θ_S (θ_D) with respect to the initial hypersurface in the Schwarzschild (de Sitter) diagram, where θ_S and θ_D are both negative (i.e., clockwise) for the case shown.

As in the real time case β_{SE} is positive (negative) if the polar angle $\tan^{-1}(V_{SE}/U_{SE})$ is increasing (decreasing), while β_{DE} is negative (positive) if the polar angle $\tan^{-1}(V_{DE}/U_{DE})$ is increasing (decreasing). The Euclidean equation of motion is the analytic continuation of (2.11), which is

$$\beta_{DE} - \beta_{SE} = \kappa r. \quad (5.13)$$

To solve this equation we re-express it as

$$\dot{r}_E^2 - V(r, M) = 1, \quad (5.14)$$

with $V(r, M)$ still given by (2.16). The Euclidean trajectory is described by motion in the potential $-V(r, M)$, with an "energy" equal to 1. The Euclidean trajectory for r is labeled (c) in Fig. 1. The signs of the β 's are also indicated in the figure. Note that β_S and β_{SE} represent the same quantity, although expressed in terms of different variables, and thus the signs shown on the figure apply to the β_E 's in the forbidden region.

Continuing to restrict ourselves to the case $M_S < M < M_{cr}$, we will describe the Euclidean geometry and evaluate the action. The bubble wall trajectory is shown on the Euclidean de Sitter and Schwarzschild spaces in Fig. 12. The initial hypersurface is labeled i in the diagram. In the de Sitter region it goes from $R = 0$ to $R = r_1$ along the $T_{DE} = 0$ line, and in the Schwarzschild region it extends from r_1 through the origin, and then onward to R_{BIG} (not shown) along the line $T_{SE} = 0$. The three-metric in this hypersurface matches exactly the three-metric on the slice i of de Sitter and Schwarzschild space shown in Fig. 11. To see this, note that the de Sitter metric (2.1) and the Euclidean de Sitter metric (5.3) are

identical when they are each restricted to surfaces of constant time; the same can be said for the Schwarzschild metric (2.6) and the Euclidean Schwarzschild metric (5.7). Since the junction between the de Sitter and Schwarzschild regions occurs at the same $R = r_1$ in both figures, the matching of the three-metrics holds throughout the hypersurface.

The final hypersurface is labeled f , and in the de Sitter region it goes from $R = 0$ to $R = r_2$ along the $T_{DE} = \bar{t}_{DE}$ line, where $\bar{t}_{DE}$ is the Euclidean de Sitter time required for the transit. From (5.12b),

$$\bar{t}_{DE} = - \int_0^{r_2} \frac{\beta_{DE}}{A_D} d\tau_E. \quad (5.15)$$

$\bar{t}_{DE}$ is related to the angle θ_D shown on the diagram by

$$\theta_D = \chi \bar{t}_{DE}. \quad (5.16)$$

By our conventions τ_E always increases along the bubble wall trajectory, so $\bar{\tau}_E > 0$. Fig. 1 shows that $\beta_D > 0$ for the case under consideration, so it follows that $\bar{t}_{DE}$ and also θ_D are negative for this case.

In the Schwarzschild region the final hypersurface goes from $R = r_2$ through the origin, and then onward to $R = R_{BIG}$ (not shown) along the line $T_{SE} = \bar{t}_{SE}$, where $\bar{t}_{SE}$ is the Euclidean Schwarzschild time required for the transit. From (5.12a),

$$\bar{t}_{SE} = \int_0^{r_2} \frac{\beta_{SE}}{A_S} d\tau_E. \quad (5.17)$$

$\bar{t}_{SE}$ is related to the angle θ_S by

$$\theta_S = \bar{t}_{SE}/4GM. \quad (5.18)$$

One can see from Fig. 1 that β_{SE} is negative for this case, so $\bar{t}_{SE}$ and θ_S are also negative. Like the initial hypersurface, the hypersurface f in Fig. 12 has a three-metric which agrees precisely with that of its counterpart in Fig. 11.

Using methods directly analogous to those used earlier, one can easily check that the initial and final hypersurfaces shown in Fig. 12 have vanishing extrinsic curvature.

From Fig. 12 we can see that the de Sitter half of the diagram has a simple interpretation. We can visualize the initial surface evolving into the final surface, while at each stage of the evolution the surface extends from the bubble wall to the edge of the de Sitter disk, where $R = 0$. The region swept out by this motion, shown as a dotted region in Fig. 13, is the de Sitter part of the interpolating four-geometry.

The Schwarzschild half of the diagram, however, is problematic. Since the Schwarzschild angle θ_S has a magnitude greater than π , the bubble wall trajectory crosses both the initial and final surfaces. In addition, the initial and final surfaces also cross each other. We have carried out rather extensive numerical calculations showing that the Schwarzschild angle θ_S always has a magnitude greater than π , so the qualitative features of the diagram are not particular to the chosen values of the parameters.

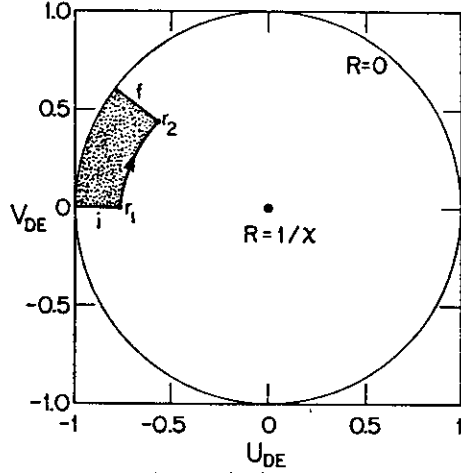


Fig. 13: The de Sitter part of the interpolating Euclidean four-geometry. The dotted area shows the region swept out as the initial hypersurface i evolves to become the final hypersurface f .

As in the de Sitter case, we can visualize the initial surface evolving into the final surface; this time, however, it is more complicated. We might imagine, for example, that the surface pivots about the origin, with one end extending to the bubble wall, and the other end to infinity. However, since some points will be crossed more than once by this motion, we cannot identify a region swept out. There is no simple analog to the dotted region shown for the de Sitter case in Fig. 13.

Nonetheless, it is possible to define an object, which we call a “pseudomanifold”, to characterize the interpolating solution. We will give two alternative descriptions of a pseudomanifold—one we will discuss here, and the other at the end of this section. The first description begins by assigning to each point a “covering number” associated with the evolution of the initial surface into the final surface. To define the covering number, we first label the two sides of the evolving surface: at the point where the surface intersects the bubble wall trajectory, we label the sides according to whether they face the *future* (larger r) or *past* of the bubble wall trajectory. Fig. 14 shows arrows that indicate the future side of both the initial and final surfaces. The covering number for a point is then defined as the number of times the point is crossed by the evolving surface with the future side first, minus the number of times it is crossed with the past side first. Note that the covering number is a topological invariant—it does not depend on the details of how the initial surface evolves into the final. If the evolution is modified by adding an extra wiggle, a given point might be covered an extra time in the positive direction; but since the initial and final positions of the surface are fixed, the point will also have to be covered an extra time in the negative direction, producing no net change. Fig. 14 shows the covering numbers for the different regions of the Euclidean Schwarzschild diagram; the covering numbers for this case take the values of $+1$, -1 , and 0 . For the de Sitter diagram, the dotted region of Fig. 13 has covering number 1 , and the rest of the space has covering number 0 .

The fact that the covering numbers take values other than 0 and 1 implies that we

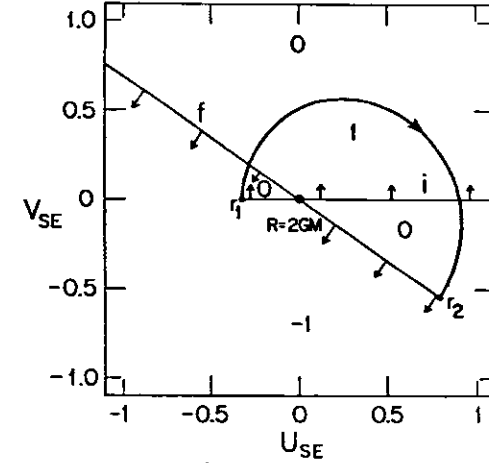


Fig. 14: Covering numbers for the Euclidean Schwarzschild space. The small arrows on the initial hypersurface i and the final hypersurface f indicate the future direction, defined by the arrow of evolution of the bubble wall trajectory at the point where the hypersurface meets the trajectory. To define the covering number, imagine the evolution of the initial hypersurface into the final. The covering number for any point is the number of times the point is crossed by the evolving surface with the future side first, minus the number of times it is crossed with the past side first.

have not found a genuine four-dimensional Euclidean manifold that interpolates between the initial and final three-geometries. We admit that we are not sure what the absence of an interpolating manifold implies about the tunneling problem. Perhaps it indicates that the stationary phase method has failed, perhaps it indicates that one cannot extrapolate the thin-wall approximation into the Euclidean regime, or perhaps it is a suggestion that tunneling is for some reason forbidden. Nonetheless, we have at least constructed a sequence of three-geometries that interpolate between the initial and final three-geometries, and perhaps this is enough to give a valid semiclassical approximation to the tunneling amplitude. For the rest of this section we will pursue the conjecture that the semiclassical method is valid in this case.

To elaborate on our conjecture, we need a procedure to evaluate the (subtracted) action of the Euclidean interpolation, described as a pseudomanifold. The action is the sum of contributions from the bubble wall, the bubble volume, the surface term, and the subtraction term: we propose a prescription in which some of these contributions are weighted by the covering number, and others are not. The prescription will also define which of the various time variables involved in this problem is to be used in evaluating the subtraction term. The prescription is:

- (1) The contribution from the bubble wall itself, including the piece arising from the matter action and the piece arising from the gravitational action, is given by integrating the analytic continuation of the Lorentzian-time expression along the bubble wall trajectory. Covering numbers are ignored, so each segment of the wall trajectory contributes once.
- (2) The bubble volume term, arising from both the matter action and the gravitational

action in the Euclidean de Sitter volume, is integrated over the volume with each region weighted by its covering number.

- (3) Surface contributions are weighted by the difference in covering number between the two sides of the surface. Surface contributions can potentially arise from the initial and final surfaces, and from the boundary at $R = R_{\text{BIG}}$, beyond which the covering number vanishes. The initial and final surfaces, however, each have zero extrinsic curvature, so the only nonzero surface contribution arises from the surface at $R = R_{\text{BIG}}$. This contribution will be canceled exactly by the subtraction of Eq. (4.20), discussed in the next paragraph, so it could be ignored from the beginning. We, however, will include the term and its canceling contribution to the subtraction.
- (4) The subtraction term of Eq. (4.20) can be described as the action of a time-independent configuration (not a solution) that exists for the same length of Euclidean time as the solution. Our problem has several time parameters, but it is the Schwarzschild time that must be used here. The reason is that Eq. (4.2), used in the derivation of (4.20), holds only for the (Euclidean) Schwarzschild time. It is the Schwarzschild time that is held fixed in the classical variational principle, and it therefore obeys the usual canonical relationships. Furthermore, we will see below that when $M < M_S$ there is a difference between the Schwarzschild time interval measured at infinity, which we denote by $\bar{t}_{SE}^\infty$, and the time interval measured along the bubble wall trajectory, which we denote by $\bar{t}_{SE}$. Since it is $\bar{t}_{SE}^\infty$ that must be held fixed in the classical variational principle, we use it to define the subtraction term.

The prescription above is motivated by the fact that it reduces to the standard Euclidean action when the space is a manifold (i.e., with covering numbers 0 and 1 only), and that its variation leads to the correct classical equations of motion (see next paragraph). Note that this prescription is a generalization of the one developed at the end of Sec. III to define the action of a region with a boundary that crosses itself.

While the trajectory shown in Figs. 13 and 14 has some properties special to the problem at hand, the action defined above can be varied to give the correct equations of motion for an arbitrary trajectory. We will demonstrate this fact at the end of this section by reformulating the prescription for the action in terms of a singular multi-sheeted representation of the pseudomanifold. That is, we will introduce a new coordinate system to describe the trajectory, with the complications of the covering numbers built into the coordinate definition. The action defined in terms of the new coordinates will have the standard form, so the derivation of the equations of motion from the action will go through in the usual way.

Since the covering number in Fig. 13 has only the values 0 and 1, the weighting by the covering number in item (2) does not affect the answer for this case. In fact, we will see below that for any value of M , the covering numbers in the de Sitter region for the relevant solutions take only the values 0 and 1. Nonetheless, the weighting by covering number would be necessary if we allowed a nonzero cosmological constant on the outside of the bubble. The weighting would also be necessary if we wished to consider a larger class of solutions to the Euclidean equations of motion. For example, if we consider a solution for which $\tau(\tau_E)$ undergoes several oscillations, then each half-cycle will be identical to the trajectory shown in Fig. 12. When the cycles are put together the trajectory will self-intersect, and then the

covering numbers would be necessary to define an action that yields the correct equations of motion.

To evaluate the subtracted action explicitly, we carry out the prescription described above. Starting with (1), the contribution of the bubble wall is the analytic continuation of the matter contribution, Eq. (3.6), and the contribution from the gravity action at the wall, Eq. (3.15). These can be added to give

$$I_E^{(1)} = \int_0^{\tau_E} \left\{ -\frac{\tau}{G} [\beta_{DE} - \beta_{SE} - \kappa\tau] + \frac{\tau^2}{2G} \left[\frac{1}{\beta_E} \left(\dot{\tau}_E - \frac{1}{2} \frac{dA}{dr} \right) \right] \right\} \Big|_S^D d\tau_E, \quad (5.19)$$

where the sign switch from Eq. (4.18) has been incorporated.

Contribution (2), the bubble volume term, has both matter and gravitational parts. In the real time case they add to give (3.9), which is ρ_f times the four-volume swept out. Since the covering number is 1 for the dotted region of Fig. 13 and 0 elsewhere, the contribution is

$$I_E^{(2)} = -\rho_f \Omega_{DE}, \quad (5.20)$$

where Ω_{DE} denotes the four-volume of the dotted region, and the sign switch again comes from (4.18). By using the (R, T_{DE}) coordinate system of Eq. (5.7), we see that

$$\Omega_{DE} = \frac{4\pi}{3} \int_0^{\tau_E} \tau^3(\tau_E) \frac{\beta_{DE}}{A_D} d\tau_E \quad (\text{for } M_S < M < M_{cr}). \quad (5.21)$$

The restriction on the mass range is shown because, as we will see later, the expression for Ω_{DE} will need to be modified when other masses are considered.

The surface contribution from $R = R_{\text{BIG}}$ is found by a derivation analogous to that of expression (3.38) for $I_{\text{net}}^{R_{\text{BIG}}}$. The resulting contribution is $M/2$ times the Euclidean Schwarzschild time interval:

$$I_E^{(3)} = \frac{M}{2} \bar{t}_{SE}^\infty, \quad (5.22)$$

where

$$\bar{t}_{SE}^\infty = \bar{t}_{SE} \quad (\text{for } M_S < M < M_{cr}), \quad (5.23)$$

and $\bar{t}_{SE}$ is given by Eq. (5.17). Here $\bar{t}_{SE}^\infty$ is the Schwarzschild time measured at infinity, while $\bar{t}_{SE}$ is the Schwarzschild time measured along the bubble wall. In the present case they are equal, but we will see that they become unequal when M falls below M_S . Eq. (5.23) will need modification, but Eq. (5.22) will remain valid. Note that $\bar{t}_{SE}^\infty/4GM$ is the angle swept out by the initial surface as it evolves into the final surface, and is not limited by 2π . Item (3) of the prescription specifies that surface terms are weighted by the covering number, so multiple coverings contribute additively. Such multiple coverings occur for small values of M/M_{cr} , and in fact the number of multiple coverings grows without bound as $M/M_{cr} \rightarrow 0$.

The subtraction term can be written as

$$I_E^{(4)} = - \left[I_E^{(1)} + I_E^{(2)} + I_E^{(3)} \right] \Big|_{\text{Static}}, \quad (5.24)$$

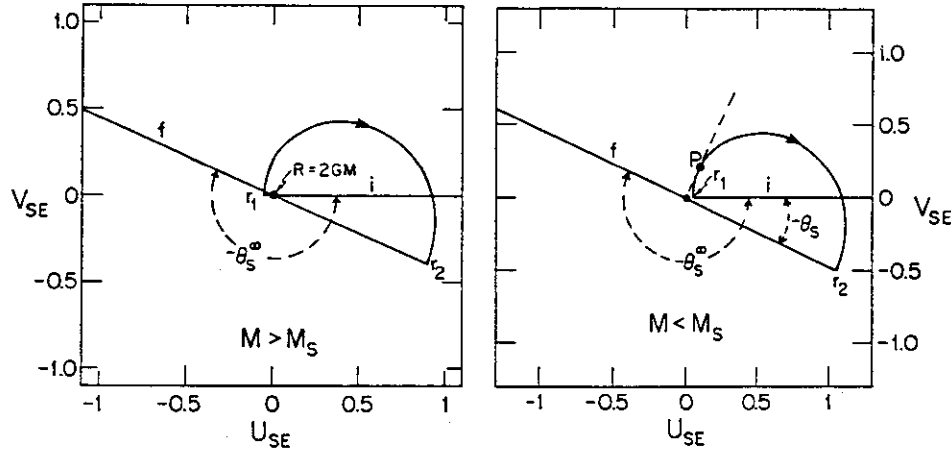


Fig. 15: The bubble wall trajectory in the Euclidean Schwarzschild space for M just above and just below M_S . The initial hypersurface is labeled i , and the final hypersurface is labeled f . The point where β_{SE} goes through zero is indicated by P . The final hypersurface is oriented at an angle θ_S^∞ with respect to the initial hypersurface. Note, however, that for $M < M_S$ the net change in the angular variable along the bubble wall trajectory is θ_S , which differs from θ_S^∞ by π . (The diagrams were computed for $\gamma = 1.5$, with $M/M_{cr} = 0.89$ and 0.93 . For comparison, $M_S/M_{cr} = 0.909$.)

where the right-hand side is evaluated for a static configuration at $r = r_1$ (the entry point to the classically forbidden region), evolving for the same Euclidean Schwarzschild time interval as the solution. Using $\beta_{SE} = -\sqrt{A_S(r_1)}$ and $\beta_{DE} = \sqrt{A_D(r_1)}$ for the static trajectory, one finds that

$$I_E^{(4)} = -M \bar{t}_{SE}^\infty. \quad (5.25)$$

Adding together the four terms, making use of the Euclidean equation of motion (5.13) and its τ_E -derivative, we finally obtain

$$I_E = - \left\{ \rho_f \Omega_{DE} + \frac{1}{2} M \bar{t}_{SE}^\infty + 2\pi\sigma \int_0^{r_B} r^2 d\tau_E \right\}. \quad (5.26)$$

We now consider what happens when the mass M is lowered through the critical value M_S . As can be seen from Fig. 1, the crucial difference is the behavior of β_{SE} . For $M > M_S$, β_{SE} is negative for the entire trajectory, while for $M < M_S$, β_{SE} starts positive and then turns negative. The form of the trajectory on the Schwarzschild diagram, for M just above and just below M_S , is shown in Fig. 15.

From the right side of Fig. 15 it can be seen that the net change in the angular variable along the bubble wall trajectory is θ_S , so

$$\theta_S = \frac{\bar{t}_{SE}}{4GM}, \quad (5.27)$$

where $\bar{t}_{SE}$ is given by Eq. (5.17). On the other hand, the angle through which the initial surface must rotate in evolving to become the final surface is θ_S^∞ , where $-\theta_S^\infty = -\theta_S + \pi$. Since $\bar{t}_{SE}^\infty$ is the Euclidean Schwarzschild time interval measured at infinity, it is proportional to θ_S^∞ :

$$\bar{t}_{SE}^\infty = 4GM\theta_S^\infty \quad (5.28a)$$

$$= \bar{t}_{SE} - 4\pi GM \Theta(M_S - M). \quad (5.28b)$$

Thus, for $M < M_S$ there is a distinction between the Schwarzschild time interval measured at infinity, $\bar{t}_{SE}^\infty$, and the Schwarzschild time interval measured along the bubble wall, $\bar{t}_{SE}$.

At first sight, the Θ -function in Eq. (5.28b) suggests that $\bar{t}_{SE}^\infty$ changes discontinuously at $M = M_S$. One sees from Fig. 15, however, that in fact θ_S^∞ is continuous at $M = M_S$, and therefore so is $\bar{t}_{SE}^\infty$. It is actually θ_S and $\bar{t}_{SE}$ that are discontinuous. In expression (5.17) for $\bar{t}_{SE}$, the origin of the discontinuity is the singular behavior of the denominator A_S . A_S is always positive in the range of the Euclidean evolution, for which $r > 2GM$, but for $M = M_S$ it vanishes at the endpoint r_1 . β_{SE} also vanishes at the endpoint for $M = M_S$, and the ratio β_{SE}/A_S is finite as $r \rightarrow r_1$. For $M = M_S \pm \epsilon$, however, the ratio β_{SE}/A_S can become arbitrarily large in the vicinity of the endpoint, and the resulting contribution to the integral is discontinuous in M . In Appendix F it is shown algebraically that the discontinuity in the integral of Eq. (5.17) is exactly what is needed so that $\bar{t}_{SE}^\infty$ has no discontinuity at $M = M_S$.

When the subtracted action is evaluated for $M_D < M < M_S$, using the prescription given above, Eq. (5.26) remains valid. The only difference is that now Eq. (5.28b) must be used to evaluate $\bar{t}_{SE}^\infty$.

Now that the distinction between $\bar{t}_{SE}$ and $\bar{t}_{SE}^\infty$ has been illustrated, we can discuss why we chose to use $\bar{t}_{SE}^\infty$ in our prescription for the subtracted action. For the surface term of item (3), the choice seems unambiguous: the boundary is located at $R = R_{BIG}$, which is equivalent to $R = \infty$, so clearly it is the time interval at infinity that determines the surface area. For the subtraction term of item (4), our choice is motivated initially by the classical variational principle, which requires $\bar{t}_{SE}^\infty$ to be held fixed. In addition, the use of $\bar{t}_{SE}^\infty$ leads to an answer that varies continuously with M , while the answer would be discontinuous if we used $\bar{t}_{SE}$. Furthermore, although we have not derived the subtraction term by a geometric argument, we would nonetheless expect the correct answer to have a geometric interpretation. The bubble wall has no geometric significance, since it is merely the place where a scalar field changes value. The Schwarzschild time interval measured at infinity, on the other hand, is part of the geometric description of the boundary (at $R = R_{BIG}$) of the spacetime region. In fact, if we had chosen to speak of a round trip "bounce" rather than a one-way transit from initial to final states, we could have described item (4) as the action of a static configuration which has the same spacetime boundary as the bounce solution. Finally, note that by choosing $\bar{t}_{SE}^\infty$ to define the subtraction of item (4), we get a cancellation of the surface term in item (3). Since the surface term has some degree of arbitrariness in its definition, we consider this an attractive property.

We now consider what happens when M is lowered below M_D . As can be seen from Fig. 1, the difference this time is in the behavior of β_{DE} . For $M > M_D$, β_{DE} is positive

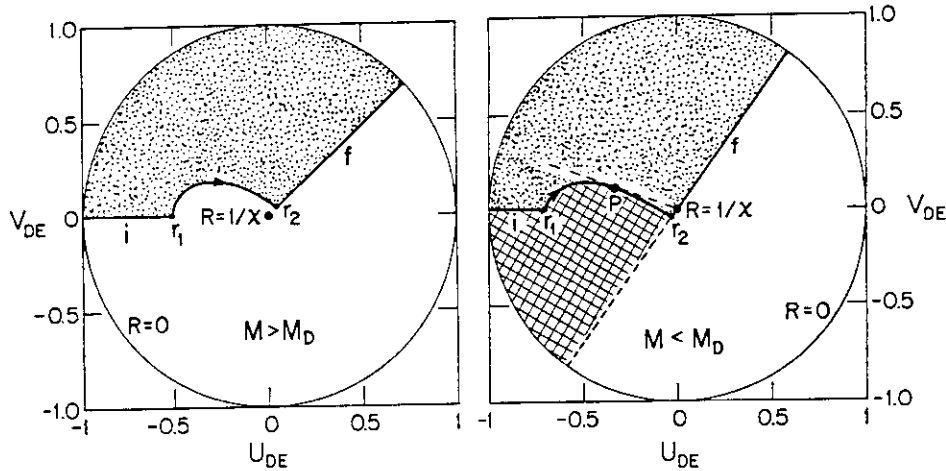


Fig. 16: The bubble wall trajectory in the Euclidean de Sitter space for M just above and just below M_D . The initial hypersurface is labeled i , and the final hypersurface is labeled f . The point where β_{DE} goes through zero is indicated by P . In each diagram the dotted region indicates the volume swept out as the initial surface evolves into the final. For $M > M_D$, the four-volume of the dotted region can be evaluated by the integral of Eq. (5.21). For $M < M_D$, however, the same integral gives the negative of the four-volume of the region indicated by cross-hatching. (The diagrams were computed for $\gamma = 1.25$, with $M/M_{cr} = 0.23$ and 0.63 . For comparison, $M_S/M_{cr} = 0.430$.)

for the entire trajectory, while for $M < M_D$, β_{DE} starts positive and then turns negative. The form of the trajectory on the de Sitter diagram, for M just above and just below M_D , is shown in Fig. 16.

To evaluate $I_E^{(2)}$, we need the four-volume Ω_{DE} of the region swept out as the initial surface evolves into the final surface, shown as a dotted region in Fig. 16. For $M > M_D$, this four-volume is given by the integral in Eq. (5.21). For $M < M_D$, however, the same integral gives the negative of the four-volume of the region indicated by cross-hatching in the diagram. To see this, note that the differential element $(\beta_{DE}/A_D) d\tau_E = -\chi^{-1} d\theta$, where θ is the polar angle of the diagram, and that the integrand represents the four-volume of the region between the curve $\tau(\tau_E)$ and the perimeter of the de Sitter disk. On the right half of Fig. 16, one obtains positive contributions until the trajectory reaches P . Then θ begins to evolve in the opposite direction, and the contribution turns negative. The initial positive contribution is canceled, and one is left with the negative contribution associated with the cross-hatched region. Thus, the four-volume Ω_{DE} of the dotted region can be written as

$$\Omega_{DE} = \frac{4\pi}{3} \int_0^{\tau_E} \tau^3(\tau_E) \frac{\beta_{DE}}{A_D} d\tau_E + \frac{4\pi^2}{3\chi^4} \Theta(M_D - M), \quad (5.29)$$

where the correction term on the right represents the four-volume of half of the full Euclidean de Sitter space.

Once again, the Θ -function appearing in Eq. (5.29) suggests that Ω_{DE} varies discontinuously with M , but in fact Ω_{DE} is continuous. For reasons parallel to those in the previous

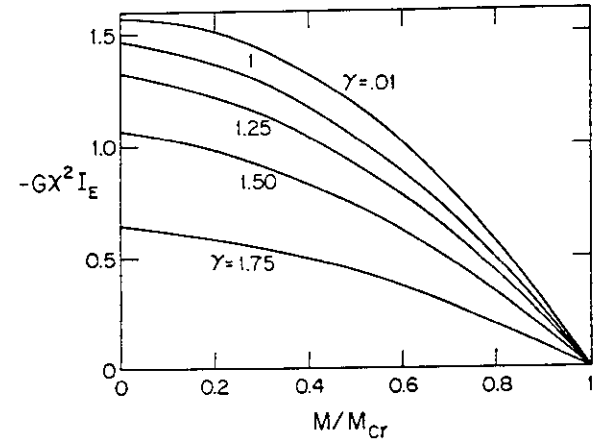


Fig. 17: Graph of the tunneling action $-G\chi^2 I_E$ as a function of M/M_{cr} , shown for various values of the parameter γ . The tunneling action for this graph was computed using the functional integral prescription, Eq. (5.30).

case, the integral in Eq. (5.29) varies discontinuously with M . The discontinuity precisely cancels the discontinuity of the Θ -function.

In evaluating the subtracted action, again Eq. (5.26) remains valid. In this case, however, we must use Eq. (5.28b) to evaluate $\bar{t}_{SE}^\infty$, and also Eq. (5.29) to evaluate Ω_{DE} . Putting all of these expressions together and making some changes of variables, one arrives at the following formula for the subtracted action, valid for any $M < M_{cr}$:

$$I_E = -\frac{1}{2G} \int_0^{\tau_E} \left\{ \kappa \tau^2 + \chi^2 \tau^3 \frac{\beta_{DE}}{A_D} + GM \frac{\beta_{SE}}{A_S} \right\} d\tau_E + 2\pi GM^2 \Theta(M_S - M) - \frac{\pi}{2G\chi^2} \Theta(M_D - M). \quad (5.30)$$

I_E is a continuous function of M , with the Θ -functions serving to cancel discontinuities that arise from the integrals.

In Fig. 17 we plot the tunneling action I_E as a function of M/M_{cr} for various values of the parameter γ , defined by Eq. (2.18). Note that I_E always has the same sign (negative), even though Eq. (5.30) contains terms of both signs. This fact is crucial to the acceptability of the answer, since a zero of I_E would correspond to the highly implausible situation of unsuppressed tunneling. Below we will give an analytic demonstration that I_E is always negative. Fig. 17 shows that the tunneling action behaves monotonically with M , and approaches zero as $M \rightarrow M_{cr}$, when tunneling is no longer necessary. Furthermore the action diverges as $G \rightarrow 0$, as one would expect. The $G \rightarrow 0$ limit can be taken with GM , χ , and κ fixed, in which case the classical trajectory remains fixed and I_E diverges as $1/G$. Alternatively one might take the limit as the physical parameters M , ρ_f , and σ are held fixed, in which case M_{cr} diverges as $1/G^{3/2}$, and I_E diverges as $1/\sqrt{G}$.

The fact that the action is negative will be viewed by some as an indication that something is wrong, since it is frequently said that the tunneling amplitude is proportional to $\exp(-I_E/\hbar)$, with no choice of signs. We argued, however, at the end of Sec. IV, that the Schrödinger equation for a tunneling process contains both exponentially growing and decaying solutions in the forbidden region—the same Schrödinger equation describes barrier penetration in either direction. We believe that the solution which falls exponentially in a given direction is always the solution that describes tunneling in that direction, independent of the sign of the Euclidean action.

To demonstrate that I_E is negative definite, we rewrite Eq. (5.30) by adding zero in a form analogous to the surface term in the action, Eq. (3.33). Consider the quantity

$$\Delta I_E \equiv -\frac{1}{2G} \int_0^{\tau_B} \frac{d}{d\tau_E} \left[r^2 \tan^{-1} \left(\frac{\dot{\tau}_E}{\beta_E} \right) \right] \Big|_S^D d\tau_E. \quad (5.31)$$

Although $\dot{\tau}_E$ vanishes at the endpoints of the integration, ΔI_E does not in general vanish. One must be careful to define the branch of the inverse tangent to be used, since the inverse tangent of zero can be any integer multiple of π . We will choose the branches:

$$\begin{aligned} -\pi &< \tan^{-1} \left(\frac{\dot{\tau}_E}{\beta_{SE}} \right) < 0, \\ 0 &< \tan^{-1} \left(\frac{\dot{\tau}_E}{\beta_{DE}} \right) < \pi. \end{aligned} \quad (5.32)$$

For this choice the inverse tangent varies continuously over the Euclidean trajectory. For $M > M_S$, $\tan^{-1}(\dot{\tau}_E/\beta_{SE})$ varies from 0 back to 0, while for $M < M_S$ it varies from $-\pi$ to 0. Similarly $\tan^{-1}(\dot{\tau}_E/\beta_{DE})$ varies from 0 to 0 for $M > M_D$, while it varies from 0 to π if $M < M_D$. (Note that the standard choice of branch, $-\frac{\pi}{2} < \tan^{-1} \theta < \frac{\pi}{2}$, would lead to discontinuities when β_{SE} or β_{DE} vanishes.) Evaluating the right-hand side of Eq. (5.31),

$$\Delta I_E = \frac{\pi \tau_1^2}{2G} \Theta(M_S - M) - \frac{\pi \tau_2^2}{2G} \Theta(M_D - M). \quad (5.33)$$

Adding ΔI_E to the right-hand side of Eq. (5.30) and subtracting the terms on the right-hand side of Eq. (5.33) to compensate, one finds after some algebra that

$$\begin{aligned} I_E = & -\frac{1}{G} \int_0^{\tau_B} r \dot{\tau}_E \left[\tan^{-1}(\dot{\tau}_E/\beta_{DE}) - \tan^{-1}(\dot{\tau}_E/\beta_{SE}) \right] d\tau_E \\ & - \frac{\pi}{2G} (\tau_1^2 - 4G^2 M^2) \Theta(M_S - M) - \frac{\pi}{2G} \left(\frac{1}{\chi^2} - \tau_2^2 \right) \Theta(M_D - M). \end{aligned} \quad (5.34)$$

Since the integrand is guaranteed to be positive by Eqs. (5.32), and since the Euclidean range for r is restricted by $2GM < r < 1/\chi$, it can be seen that each of the three terms in this expression is negative definite.

Having completed the calculation, we want to emphasize that we have not constructed an interpolating four-dimensional manifold with a well-defined scalar field, and that our

calculation of the tunneling action is therefore based on a conjecture rather than a derivation. The difficulties with the conjecture become most apparent if we try to imagine moving slightly away from the thin-wall limit, so that the scalar field is treated as a continuous function that varies quickly from ϕ_t (true vacuum value) to ϕ_f (false vacuum value) at the bubble wall. One difficulty is the inconsistency between items (1) and (2) of the prescription: the action associated with the bubble wall is to be counted only once, while the action associated with the false vacuum volume is to be weighted by the covering number. But if the field is continuous, there is no sharp distinction between these two contributions to the action. A second difficulty concerns the consistency of the scalar field function, particularly where the bubble wall trajectory intersects the initial or final surface, as shown for example in Fig. 14. If we think of these regions as part of the initial or final surface, then (in the Schwarzschild region) the value of ϕ in this vicinity should be very near ϕ_t . On the other hand, if we think of the region as being at the bubble wall, then ϕ should be varying rapidly between ϕ_t and ϕ_f .

The possibility of a multi-sheeted covering immediately comes to mind, but the possibility of a nonsingular multi-sheeted covering is excluded by the fact that the underlying space is simply connected. To have a multi-sheeted structure, there must be a rule that allows us to determine, for any closed path, whether the path returns to the original sheet or a different one. If all closed paths can be continuously deformed into a trivial path, as is the case here, then all closed paths must return to the original sheet.

Although a nonsingular multi-sheeted covering is excluded, a new coordinate system giving a *singular* multi-sheeted covering can nonetheless be constructed. The argument above can be bypassed, provided that the coordinates of the covering are not required to be continuous functions of the coordinates of the original Euclidean Schwarzschild and de Sitter manifolds. Then, as a path is continuously deformed in the Schwarzschild or de Sitter manifold, the image of the path in the new coordinates can jump discontinuously from one sheet to another.

In fact, all of the pathologies described two paragraphs above can be alleviated by redescribing the pseudomanifold by a singular multi-sheeted covering. The complications of the covering numbers can be built into this coordinate definition, so that the action has the standard form when it is expressed in the new coordinates. The metric in terms of these new coordinates is singular, in the sense that $\sqrt{g}$ can vanish and change sign, but the gravitational and scalar field equations are satisfied everywhere.

To construct the new coordinate system, $\xi^\mu \equiv (\tau_E, \sigma, \theta, \phi)$, we first parameterize the bubble wall trajectory, on both the de Sitter and Schwarzschild diagrams, by the proper time variable τ_E . Then for each point $P(\tau_E)$ along the trajectory we choose a surface that extends from $R = 0$ in the de Sitter region to the point P , and then in the Schwarzschild region from P to $R = \infty$. (There is no requirement, however, that R vary monotonically along this surface.) We further insist that the surfaces are chosen continuously in τ_E , and that the initial and final surfaces coincide with the initial and final surfaces of the tunneling problem. We then define on each surface a coordinate σ , extending from 0 at $R = 0$ to ∞ at $R = \infty$, insisting that σ is continuous in τ_E and that it varies continuously along the surface. We let $\sigma(\tau_E)$ indicate the value of the σ -coordinate at the point P . Note that some points in the original Schwarzschild and de Sitter manifolds will be covered several times by

this procedure, and other points will not be covered at all— if a sign is attributed to each covering of this evolving surface as before, then the net number of times a point is covered is just the covering number defined earlier.

We now assign a metric $\tilde{g}_{\mu\nu}$ to the ξ -space as follows:

$$\tilde{g}_{\mu\nu}(\xi) = \begin{cases} \frac{\partial x^\lambda}{\partial \xi^\mu} \frac{\partial x^\rho}{\partial \xi^\nu} g_{\lambda\rho}^{DE}(x(\xi)) & \text{if } \sigma < \sigma(\tau_E) \\ \frac{\partial x^\lambda}{\partial \xi^\mu} \frac{\partial x^\rho}{\partial \xi^\nu} g_{\lambda\rho}^{SE}(x(\xi)) & \text{if } \sigma > \sigma(\tau_E) . \end{cases} \quad (5.35)$$

where $g_{\lambda\rho}^{DE}$ and $g_{\lambda\rho}^{SE}$ denote the Euclidean De Sitter and Schwarzschild metrics, respectively. Note that on each slice the metric has only a single de Sitter region ($\sigma < \sigma(\tau_E)$) and a single Schwarzschild region ($\sigma > \sigma(\tau_E)$), even if the surface crosses the domain wall in Figs. 13 and 14 many times. Note also that the junction conditions already imposed on the domain wall trajectory imply that the metric $\tilde{g}_{\mu\nu}$ describes a space that has no discontinuity at $\sigma = \sigma(\tau_E)$. As in Sec. III, we could use Gaussian normal coordinates to describe the space in a manner which is manifestly continuous in the vicinity of the wall.

In the modified coordinate system the action defined by the prescription given earlier in this section can be written simply as

$$I_E = \int d^4\xi \sqrt{\tilde{g}} [\mathcal{L}_{\text{matter}}^E + \mathcal{L}_{\text{gravity}}^E] + \frac{1}{8\pi G} \int d^3\xi \sqrt{\tilde{h}} K , \quad (5.36)$$

where $\mathcal{L}_{\text{matter}}^E$ and $\mathcal{L}_{\text{gravity}}^E$ are the usual Euclideanized expressions for the matter and gravity Lagrangians, K is the extrinsic curvature of the boundary, and

$$\sqrt{\tilde{g}(\xi)} \equiv \det \left(\frac{\partial x}{\partial \xi} \right) \sqrt{g(x)} \quad (5.37)$$

is allowed to have either sign, depending on the sign of $\det(\partial x / \partial \xi)$. Note that the ξ -space is not a Riemannian manifold, since the determinant of the metric can vanish and change sign.

As we mentioned earlier this singular multi-sheeted representation of the pseudomanifold can be used to show that the variation of the action leads to the correct equations of motion. To see this, note that Eq. (5.36) is algebraically identical to the expression for the action defined on a manifold, and that the formal manipulations involved in varying the action to obtain the equations of motion do not depend on the sign of $\sqrt{\tilde{g}}$. Since we know that the action defined on a manifold leads to the correct equations of motion, it follows that the action defined above will do the same.

The singular multi-sheeted covering has the advantages that the expression for the action has the usual form, the scalar field is single-valued, the field equations hold everywhere, and there is nothing that prevents the wall from acquiring a small thickness. These advantages are attained, however, at the price of introducing a new pathology: the metric in the new coordinates is singular, in the sense that $\sqrt{\tilde{g}}$ can vanish and change sign. Despite this pathology, it is still conceivable that metrics of this type are included in the functional integral for quantum gravity. Alternatively, even if such metrics are not included, it is still possible that they arise as saddle points which are obtained by the distortion of integration contours in the complex plane.

VI. CANONICAL QUANTIZATION OF THE THIN-WALL BUBBLE

In this section we describe an attempt to canonically quantize the thin-wall bubble Lagrangian, Eq. (3.47). We find, however, that the canonical procedure breaks down at an early stage: the relation $p(r, dr/dt_S)$ is noninvertible, so the canonical momentum p does not uniquely determine the coordinate velocity $r' \equiv dr/dt_S$. We nonetheless try to ignore the problem and go on, motivated by the belief that anyone wishing to explore variations of the canonical method should be aware of all the pitfalls that have to be circumvented. In particular, we find that the canonical momentum cannot in general be defined so that it vanishes at both endpoints of the tunneling, and we also find that the numerical answer varies non-monotonically with M/M_{cr} , which we find difficult to believe.

Starting with the Lagrangian $L_S(r, r')$ given by Eqs. (3.46) and (3.47), one finds that the canonical momentum is given by

$$p = \frac{\partial L_S}{\partial r'} = \frac{r}{G} \left[\tanh^{-1}(\dot{r}/\beta_S) - \tanh^{-1}(\dot{r}/\beta_D) \right] - \frac{r\dot{r}}{G\beta_S} [\beta_D - \beta_S - \kappa r] , \quad (6.1)$$

where again $\kappa \equiv 4\pi G\sigma$. The resulting Hamiltonian is

$$H = M - \frac{r\beta_S}{G} [\beta_D - \beta_S - \kappa r] . \quad (6.2)$$

Note that the last terms in each of the above equations vanish when the equations of motion (2.11) are satisfied, so in particular the constant value of H is given by M .

To continue explicitly we would have to invert Eq. (6.1) to express $\dot{r}$ in terms of p , so that we could then write the Hamiltonian in terms of p and r . (The canonical velocity is really r' , but $\dot{r}$ and r' are related by Eq. (3.45), and for many purposes $\dot{r}$ is the simpler quantity to deal with). At this point, however, we find that the procedure fails. The problem is that Eq. (6.1) describes a noninvertible function. To see this, we calculate

$$\frac{\partial p}{\partial \dot{r}} = \frac{r}{G\beta_S\beta_D} [\kappa r\beta_D - (\beta_D - \beta_S)^2] . \quad (6.3)$$

The expression on the right can have either sign, so p is not a monotonic function of $\dot{r}$. (If the reader has any doubt about this, she can evaluate the expression for a trajectory — not necessarily a solution — in the right quadrant of the Kruskal diagram and the left quadrant of the de Sitter diagram, for a value of r infinitesimally less than $1/\chi$. For $\dot{r} = 0$ one finds $\partial p / \partial \dot{r} \approx -r/G\sqrt{\beta_D} < 0$, while for $\dot{r} \rightarrow \infty$ one finds $\partial p / \partial \dot{r} \approx \kappa r^2 / G\beta_S > 0$.)

Since the Hamiltonian is not a well-defined function of p and r , our program of canonical quantization is clearly derailed. Nonetheless, the tunneling amplitude expressed by Eqs. (4.20) and (4.21) does not require an explicit form for the Hamiltonian. It is therefore tempting to proceed, but we must remember that the calculation is not justified.

The Euclidean action I_E of Eq. (4.20) is expressed as an integral over t_{SE} , but it can be re-expressed as an integral over the Euclidean proper time variable. Using Eqs. (4.14), (3.47), (5.12a), and the equation of motion (5.13), we find

$$I_E^{\text{can}} = \frac{1}{G} \int_0^{\tau_E} \dot{r}_E \left[\tan^{-1}(\dot{r}_E/\beta_{SE}) - \tan^{-1}(\dot{r}_E/\beta_{DE}) \right] d\tau_E. \quad (6.4)$$

As in the previous section, $r(\tau_E)$ is the solution of the Euclidean equation of motion that evolves from r_1 to r_2 , and τ_E is the Euclidean proper time for the transit. In this form I_E^{can} can easily be seen to agree with the general result $I_E = -i \int p(dr/d\tau_E) d\tau_E$, where p is given by the analytic continuation of Eq. (6.1), the second term of which vanishes by the equation of motion (5.13).

To evaluate (6.4) one must first specify the appropriate branch of the inverse tangent function. In making this choice, one must recognize that the WKB wave function of Eqs. (4.10) and (4.13) will approximately satisfy the Schrödinger equation only if $p(r)$ is defined continuously. As an example one can consider a free nonrelativistic particle in one dimension, for which $p = \pm\sqrt{2mE}$; either root can be used, but the Schrödinger equation is not even approximately satisfied if p jumps discontinuously from one branch to the other. Thus, the root of the inverse tangent function must be chosen continuously in r .

Since either β_E can reverse its sign during the Euclidean evolution, while $\dot{r}_E$ is positive, the argument $\dot{r}_E/\beta_E$ can go through infinity but never zero. Thus if the branches are chosen according to Eq. (5.32), then the inverse tangents will vary continuously over the Euclidean trajectory. As discussed in Sec. V, $\tan^{-1}(\dot{r}_E/\beta_{SE})$ varies from 0 back to 0 for $M > M_S$, while for $M < M_S$ it varies from $-\pi$ to 0. Similarly $\tan^{-1}(\dot{r}_E/\beta_{DE})$ varies from 0 to 0 for $M > M_D$, while it varies from 0 to π if $M < M_D$. This choice is by no means unique, but other choices will simply change the quantity in square brackets by $N\pi$ (N an integer), resulting in a change in the action by

$$\Delta I_E^{\text{can}} = \frac{N\pi}{2G} (r_2^2 - r_1^2). \quad (6.5)$$

Note that for $M_D < M < M_S$, the quantity in brackets in Eq. (6.4) increases by π during the Euclidean evolution. Thus it cannot vanish at both endpoints no matter what branches are chosen, and therefore neither can the canonical momentum.

To compare with the geometric answer of Sec. V, it is easiest to refer to expression (5.34). Note that I_E^{can} matches identically the first term of (5.34). Thus, for $M > M_S$, the two answers agree. For $M_D < M < M_S$, however, the geometric answer is modified by the second term of (5.34):

$$-\frac{\pi}{2G} (r_1^2 - 4G^2 M^2) \Theta(M_S - M). \quad (6.6)$$

The discrepancy between the two results can be understood in geometric terms, and is related to the slicing prescription used in Sec. III. The canonical formulation used here associates each point along the bubble wall trajectory with a hypersurface of constant Schwarzschild time T_S . After Euclideanization, such a hypersurface corresponds to a radial

line in the Kruskal diagram. From the left of Fig. 15, we can see that for $M > M_S$ the canonical prescription agrees with the geometric prescription. Each canonical hypersurface is a straight line extending from a point on the wall trajectory through the origin and onward to infinity. These hypersurfaces evolve continuously from the initial hypersurface i to the final hypersurface f , just as was assumed in the geometric description of Sec. V. For $M < M_S$, however, the canonical procedure applies the same slicing prescription to the right half of Fig. 15. In this case the initial hypersurface of the canonical procedure does not agree with i ; instead it starts at the bubble wall and extends through the origin and onward to the left. At each stage of evolution the canonical hypersurface is a radial line from the bubble wall through the origin and onward to infinity, so it evolves smoothly into the final hypersurface f .

The fact that for $M < M_S$ the canonical approach has the initial surface pointing in the wrong direction explains why the initial value of $\tan^{-1}(\dot{r}_E/\beta_{SE})$ is nonzero. As shown in Sec. III, the inverse tangent is a measure of the angle of the initial surface, which the canonical procedure has wrong by an angle π . This angle produces a change in the surface term of the action, Eq. (3.32), which accounts for the discrepancy of $\pi r_1^2/2G$ between the two answers. (Note that the surface term is making a nonzero contribution to the canonical answer; we consider this yet another sign that the canonical answer is flawed.) The misdirection of the initial surface also produces a discrepancy of π in the total Schwarzschild angle of the solution. Through the surface term at $R = R_{BIG}$ and the subtraction term, this allows us to understand the remaining discrepancy shown in expression (6.6).

If the mass is lowered below M_D , then β_{DE} will also change sign during the Euclidean evolution. An additional discrepancy between the canonical answer and the geometric answer arises from the third term of (5.34), and again the discrepancy can be understood in terms of the geometry of the canonical recipe for spacetime slicing. In this case it is the orientation of the final slice in the de Sitter region that is wrong by an angle of π .

The physical significance of these misoriented initial and final hypersurfaces is probably best understood by considering again the WKB wave function. When the momentum p is nonzero at one or both endpoints, continuity implies that the inverse hyperbolic tangent in the neighboring allowed region must be defined to include an imaginary part equal to π . This means that the wave function under discussion varies exponentially with r , even in the allowed region. Clearly the canonical formulation is not giving us what we expect. The problems with the canonical formulation are apparently due, at least in part, to the inadequacy of our recipe for choosing spacelike hypersurfaces. The problem shows up algebraically in the absence of a one to one relationship between $\dot{r}$ and p . While the classical solution has the property that $\dot{r} \rightarrow 0$ as $r \rightarrow r_1$ or $r \rightarrow r_2$, it does not follow that $p \rightarrow 0$.

Finally, we ask for curiosity how the numerical value of the canonical answer behaves. The result is shown in Fig. 18. Note that I_E^{can} does not vary monotonically with M/M_{cr} , as one expects that the correct answer must. Furthermore, we have numerically explored the possibility of other choices of branch, finding that monotonicity in M/M_{cr} cannot be obtained for any choice. The problem is that ΔI_E^{can} is not monotonic in M/M_{cr} , and there are ranges of parameters for which its non-monotonicity dominates for any nonzero integer N .

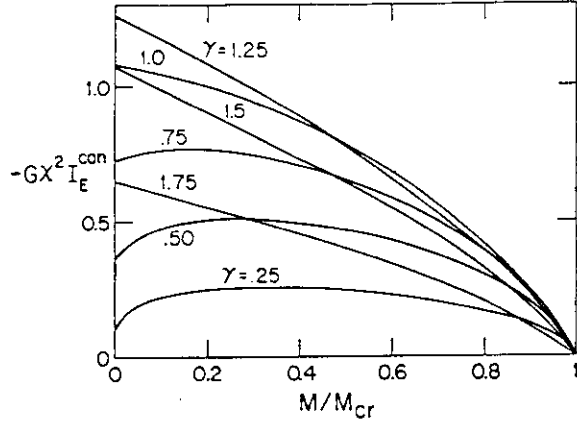


Fig. 18: Graph of the tunneling action $-GX^2 I_E^{\text{con}}$ as a function of M/M_{cr} , shown for various values of the parameter γ . This graph is based on the canonical result Eq. (6.4), which we do not believe is valid.

VII. CONCLUSIONS

The inflationary universe model proposes that our universe grew from a tiny inflating region of false vacuum. We know, however, that the laws of classical general relativity imply that a bubble that grows large enough to become a new universe cannot be produced without an initial singularity. In this paper we have asked whether this requirement can be avoided by quantum tunneling. Unfortunately we do not have a definitive answer to this question, but we have obtained an expression for the tunneling amplitude that seems highly plausible, and we conjecture that it is a valid approximation. Thus, we are suggesting that quantum effects can very likely avoid the implications of the classical singularity theorems, and that the laws of physics as we know them permit in principle the creation of a new universe by human initiative. In addressing this question, however, we have uncovered a situation in the semiclassical formulation of quantum gravity that appears very novel: we have found a tunneling process for which the interpolating Euclidean solution folds back on itself in a manner that cannot be interpreted as a manifold. To describe this object we have introduced what we call a pseudomanifold, a generalization of a manifold in which each point is assigned an integer-valued covering number. The pseudomanifold can also be described by a multi-sheeted covering that is singular in the sense that $\sqrt{g}$, the square root of the determinant of the metric, can vanish and change sign.

The starting point of our investigation was the observation that in the classical thin-wall bubble problem there can be two very different types of trajectories with the same mass M , provided that M is less than a critical mass M_{cr} . One type corresponds to a bubble which, with some fancy engineering, could in principle be constructed. In fact for M very small, gravity plays no role and the bubble can be viewed as a conventional flat space false vacuum bubble. According to classical physics, this bubble would expand to a maximum radius and then collapse. The second corresponds to a bubble which grows arbitrarily large, but which,

according to classical general relativity, could not be produced without an initial singularity. Since the two configurations have the same mass, it is natural to consider the possibility that the first configuration could tunnel to become the second.

Our experience with the WKB method has taught us that Euclidean solutions to classical equations of motion can be used to estimate tunneling rates. If we reduce our problem to just one degree of freedom, the radius of the bubble wall, we can then find an imaginary time solution which goes through the classically forbidden region. This solution starts from the maximum radius of the initial classical trajectory and ends at the minimum radius of the final classical trajectory. From this point of view the problem looks very much like an ordinary one-dimensional tunneling problem, with the Euclidean solution interpolating between the turning points. At this level it appears as if the problem is straightforward.

When the calculation is continued, however, complications become apparent. We have pursued both a canonical and a functional integral approach, finding unexpected results in both cases. We believe that our canonical approach is clearly invalid, but we discuss it anyway for its pedagogical value. The functional integral approach also has complications that we do not fully understand, but it nonetheless leads to a result which is plausibly correct.

In order to construct a canonical description for the one-degree-of-freedom problem, it is necessary to reformulate the action of the full field theory, which is expressed as an integral over the four-dimensional spacetime. To do this we adopted a recipe for slicing spacetime. Each segment of the trajectory was then associated with a region of spacetime, and the action for the segment was taken as the field theory action for the corresponding region. In order to maintain explicit time-translation invariance, we chose to slice along hypersurfaces of constant Schwarzschild time or static de Sitter time. This prescription, however, works only for trajectories that lie in the left or right quadrants of the Kruskal diagram and in the left or right quadrants of the de Sitter diagram.

Pursuing the canonical approach, we found that the expression for the canonical momentum $p(r, \dot{r})$ is noninvertible, so $\dot{r}$ cannot be expressed as a function of r and p . This means that it is impossible to define a Hamiltonian $H(r, p)$, and the canonical formulation cannot be completed. It is nonetheless possible to evaluate the action for the solution, and we found that the action varies non-monotonically with the bubble mass M , which seems very hard to accept. We also found that, for a range of bubble masses ($M_D < M < M_S$), the canonical momentum p cannot be defined so that it vanishes at both ends of the tunneling trajectory.

The alternative, and in this case more successful approach, is the sum over histories. While we cannot even attempt a calculation of the full functional integral, we assume that to leading order in $\hbar$ the amplitude can be approximated by semiclassical methods. That is, we assume that the amplitude to go from one three-geometry to another is well-approximated by $e^{iI_{\text{cl}}/\hbar}$ where I_{cl} is the action of the classical solution to the field equations which interpolates between the two three-geometries. If no real-time solution exists then we seek a Euclidean four-geometry that solves the imaginary time field equation and whose boundary is the two three-geometries of interest. The tunneling amplitude is then estimated as $e^{-I_E/\hbar}$, where I_E is the properly subtracted classical action of the Euclidean solution—that is, it is the action of the solution, minus the action of a configuration that remains static at the

initial state of the tunneling process for the same Euclidean time as the solution requires for its transit.

We have found, however, that no true Euclidean interpolating manifold exists. There is no difficulty or ambiguity in analytically continuing the bubble wall trajectory into the Euclidean regime, but when this trajectory is plotted on a Euclidean spacetime diagram it is found to cross both the initial and final surfaces of the tunneling problem. These intersection points prevent a conventional manifold interpretation.

We admit that we are not sure what the absence of a true interpolating manifold implies about the tunneling problem. Perhaps it indicates that the stationary phase method has failed, perhaps it indicates that one cannot extrapolate the thin-wall approximation into the Euclidean regime, or perhaps it is a suggestion that tunneling is for some reason forbidden.

We find it difficult to believe, however, that the tunneling process is forbidden, since there is no barrier to constructing a well-defined manifold (with either Lorentzian or Euclidean signature) that interpolates between the initial and final states. Such a manifold is not a solution, but it would constitute a path contributing to the functional integral. Furthermore, since any small variation about such a path would also contribute, the measure of these paths appears naively to be nonzero. The amplitude would then be nonzero unless the various paths conspire to cancel each other, as they do for an amplitude that violates a conservation principle associated with a symmetry. In the present case, however, there is no apparent symmetry or conservation law at work. We therefore conjecture that the tunneling process is allowed, and that the semiclassical approximation is valid.

To characterize the Euclidean interpolation we define a pseudomanifold, which we describe in two alternative ways. In either case, it is specified in terms of the behavior of a hypersurface that evolves continuously through the Euclidean Schwarzschild and de Sitter manifolds, beginning at the initial hypersurface of the tunneling process and ending at the final hypersurface. In one description each point is assigned a covering number, defined as the number of times the point is crossed by the evolving hypersurface, with crossings counting positively in one direction and negatively in the other. In the other description we introduce new coordinates that give a singular multi-sheeted covering of the original space. In these new coordinates it looks very much like a true manifold, except that $\sqrt{g}$ is allowed to vanish and to change sign. We have given a prescription to define the action for a pseudomanifold, which in the singular multi-sheeted description has the same form as the usual expression for the action. It is then clear that the variation of this action reproduces the (analytically continued) classical equations of motion.

We have used our definition of the action to estimate the tunneling amplitude as a function of the various parameters in the problem, and we have found that it behaves very reasonably: the tunneling action decreases monotonically to zero as the bubble mass M approaches the critical mass M_{cr} at which tunneling would not be necessary, and it diverges monotonically as the gravitational constant $G \rightarrow 0$. The action is negative definite by the usual sign conventions, but we argue that, regardless of the sign of the action, the tunneling probability is always exponentially suppressed.

In a recent preprint, Fischler, Morgan, and Polchinski¹⁵ have calculated an amplitude for this same process, using a Hamiltonian method somewhat different from the method

described in this paper. They say that they find no inconsistencies, and they obtain the same answer we find using the functional integral method.²⁴

As a rough estimate, the action I_E is of order $1/(G\chi^2)$, as long as $G\sigma^2$ is smaller than or comparable to ρ_f , and M is not too near M_{cr} . For typical grand unified theory parameters, this would give an outrageously small tunneling probability, such as $10^{-10^{16}}$. Even with this small probability, however, there might still be a large probability of an event of this sort occurring somewhere in a universe that has undergone a large amount of inflation. Thus, the possibility of a chain reaction by which one universe produces more than one universe is not obviously ruled out by this estimate. On the other hand, if we are talking about human-made universes, then a probability this small must be considered equivalent to zero. Thus the production of a universe at the GUT scale seems prohibitively unlikely, but it might be possible at energy scales approaching the Planck scale.

If our functional integral result is correct, then it seems to raise an important issue in quantum gravity: how does a pseudomanifold arise in a quantum gravity path integral? It might mean that such objects occur in the physical definition of the path integral, or it might mean that they arise as saddle points which are obtained by the distortion of integration contours in the complex plane.

Finally we would like to discuss what the production of a new universe would look like to us, the creators of the new spacetime. Recall that to produce a new universe it is necessary to build very many false vacuum bubbles, each in its expanding phase. Almost all of these bubbles will follow their classical trajectories— that is, they will expand to a maximum radius and then collapse. To an observer riding along with the wall, the wall is seen to pass through the Schwarzschild radius of the bubble in a finite time. However to an observer who sits at a fixed radius, beyond the trajectory, the wall takes an infinite time to reach the horizon. At late times the wall appears frozen at the horizon. If the observer sends in light rays to probe the bubble wall, then the reflected waves are received for all time, although they become highly redshifted.

A few of the bubbles, roughly e^{-2I_E} times the number we start with, reach their maximum classical radius and then quantum tunnel to a larger radius on the other side of the horizon. These are the seeds of the new universes. The time-energy uncertainty relation prevents us from saying exactly what this event looks like at the instant it occurs. After the tunneling event, however, the bubble trajectory closely resembles that of the classical trajectory shown in Fig. 5. Here the horizon is exposed; there is no wall frozen just outside of the horizon. If an observer at fixed radius has been illuminating the bubble wall with light rays, the reflected rays would cease during the tunneling process. When the bubble tunnels it vanishes from your radar screen.

We must ask, however, whether the vanishing of the bubble is an unmistakable signal that a new universe has been produced. Are there other processes that would look the same from the outside, but which do not lead to the creation of a new universe? To answer the question we need to go beyond the approximations used in this paper, in which the bubble motion before and after the tunneling is treated classically. We need instead a full quantum description, with the bubble motion represented by the evolution of a wave packet. The packet might be highly peaked at the classical trajectory, but the spread of the packet would

nonetheless allow the possibility of evolution very different from classical expectations. Thus, there is some probability that reflected photons would cease after a certain time, even if a new universe is not created. We have not yet attempted to estimate this probability.

The lesson is that even if a new universe can be created, we still do not know if this spectacular event could be verified.

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18. Note that Eq. (4.25a) of Ref. [6], defining β_D , contains a sign error. For a trajectory moving upward on the left side of the de Sitter diagram, t_D is negative, while β_D is positive.
19. Note that there are no closed anti-trapped surfaces once the trajectory enters the upper quadrant of the Kruskal diagram, so in principle it is possible to classically produce the bubble in this late stage of collapse. Such a bubble would immediately collapse, however, with no chance for quantum mechanical tunneling.
20. We follow the sign conventions of Ref. [25], in which the key definitions are given by $\Gamma_{\lambda\sigma}^\mu = \frac{1}{2}g^{\mu\nu}(\partial_\lambda g_{\nu\sigma} + \partial_\sigma g_{\nu\lambda} - \partial_\nu g_{\lambda\sigma})$, $V^\mu{}_{;\nu} = \partial_\nu V^\mu + \Gamma_{\nu\lambda}^\mu V^\lambda$, $V_{\mu;\nu} = \partial_\nu V_\mu - \Gamma_{\mu\nu}^\lambda V_\lambda$, $R^\mu{}_{\nu\alpha\beta} = \partial_\alpha \Gamma_{\nu\beta}^\mu - \partial_\beta \Gamma_{\nu\alpha}^\mu + \Gamma_{\sigma\alpha}^\mu \Gamma_{\nu\beta}^\sigma - \Gamma_{\sigma\beta}^\mu \Gamma_{\nu\alpha}^\sigma$, and $R_{\mu\nu} = R^\alpha{}_{\mu\alpha\nu}$.
21. G. W. Gibbons and S. W. Hawking, *Phys. Rev. D* **15**, 2752 (1977). See also S. W. Hawking, in *General Relativity: An Einstein Centenary Survey*, eds: S.W. Hawking and W. Israel (Cambridge University Press, Cambridge, 1979), p. 746.
22. To make contact with Eqs. (3.1) and (3.12), note that in Gaussian normal coordinates $(\tau, \theta, \varphi, \eta)$, the unit normal is given simply by $n^\mu = (0, 0, 0, 1)$. Then $\nabla_\mu n^\mu = g^{-1/2} \partial_\mu (g^{1/2} n^\mu) = g^{-1/2} \partial_\eta g^{1/2} = \frac{1}{2} g^{ij} \partial_\eta g_{ij}$.
23. The expression for the normal vector $n^\mu(x, y)$ can be defined arbitrarily for $y \neq f(x)$, as long as it is differentiable and properly normalized. The normalization of n^μ implies that the normal derivative can be subtracted without any effect: $\nabla_\mu n^\mu = (\nabla_\mu - n_\mu n^\nu \nabla_\nu) n^\mu$.
24. To compare the answer of Ref. [15] with ours, it is easiest to refer to Eq. (5.34). Using trigonometric identities and the equations of motion, one can see that the quantity in square brackets in the integrand of (5.34) is equal to π minus the inverse cosine appearing in Eq. (11) of Ref. [15]. Note also that the sign of the integral in Eq. (11) of Ref. [15] was misprinted in the early version of the preprint, but should be negative. The result in the preprint is limited to the range $M_D < M < M_S$, but the authors have since extended their calculations to arbitrary M , continuing to agree with Eq. (5.34). We thank J. Polchinski for informing us of these results prior to their publication.
25. C. W. Misner, K. S. Thorne and J. A. Wheeler, *Gravitation* (W. H. Freeman and Company, San Francisco, 1973).

APPENDIX A GAUSSIAN NORMAL COORDINATES

In this appendix we construct exact expressions for the transformation from static coordinates to the Gaussian normal coordinate system based on the bubble wall trajectory. Although our calculations make use of these expressions only to first order in the distance η from the wall, the exact expressions are so simple that we feel it is worthwhile to present them. We will not, however, attempt to determine how much of the manifold is covered by the Gaussian normal system.

In static coordinates, the metric is given on the Schwarzschild side by Eq. (2.1) and on the de Sitter side by Eq. (2.6). In this appendix we will suppress the subscripts on A and T appearing in these expressions, so that the Schwarzschild and de Sitter cases can be treated at once. Our goal is to transform the static coordinates (T, R) to the Gaussian normal coordinates (τ, η) , with the metric given by Eq. (3.5). The requirement that the Gaussian normal coordinates be centered on the bubble wall trajectory implies that

$$R(\tau, \eta=0) = r(\tau) . \quad (\text{A.1})$$

The transformation equations for the metric components give

$$g_{\eta\eta} = 1 = -A(R) \left(\frac{\partial T}{\partial \eta} \right)^2 + \frac{1}{A(R)} \left(\frac{\partial R}{\partial \eta} \right)^2 \quad (\text{A.2})$$

$$g_{\tau\eta} = 0 = -A(R) \frac{\partial T}{\partial \eta} \frac{\partial T}{\partial \tau} + \frac{1}{A(R)} \frac{\partial R}{\partial \eta} \frac{\partial R}{\partial \tau} \quad (\text{A.3})$$

$$g_{\tau\tau}(\tau, \eta) = -A(R) \left(\frac{\partial T}{\partial \tau} \right)^2 + \frac{1}{A(R)} \left(\frac{\partial R}{\partial \tau} \right)^2 , \quad (\text{A.4})$$

where

$$g_{\tau\tau}(\tau, 0) = -1 . \quad (\text{A.5})$$

(For completeness, we point out that the vanishing of $g_{\tau\eta}$ for all η follows from writing the geodesic equations, in the Gaussian normal system, for the curve generated by varying η with τ , θ , and ϕ held fixed.)

At $\eta = 0$, where Eq. (A.5) applies, Eqs. (A.2-5) provide three equations for the four unknown partial derivatives of the coordinate transformation. Differentiation of (A.1) provides the necessary fourth equation, yielding

$$\frac{\partial R}{\partial \tau} = \dot{r}(\tau) \quad (\text{A.6a})$$

$$\frac{\partial R}{\partial \eta} = \beta(r(\tau), \dot{r}(\tau)) \quad (\text{A.6b})$$

$$\frac{\partial T}{\partial \tau} = \pm \beta(r(\tau), \dot{r}(\tau)) / A(r(\tau)) \quad (\text{A.6c})$$

and

$$\frac{\partial T}{\partial \eta} = \pm \dot{r}(\tau) / A(r(\tau)) , \quad (\text{A.6d})$$

all at $\eta = 0$, where β is defined by Eq. (2.14) with the subsequent discussion of its sign, and the upper and lower signs of Eqs. (A.6c-d) refer to Schwarzschild and de Sitter space, respectively. Note that the sign choice in Eq. (A.6b) is the mathematical statement of our sign convention for η .

To extend the equations to $\eta \neq 0$, use the fact that the curve generated by varying η , with τ , θ , and ϕ held fixed, is by construction a geodesic. The geodesic equations then give

$$\frac{\partial}{\partial \eta} \left[A(R) \frac{\partial T}{\partial \eta} \right] = 0 \quad (\text{A.7})$$

and

$$\frac{\partial^2 R}{\partial \eta^2} - \frac{1}{2} \frac{dA}{dR} = 0 . \quad (\text{A.8})$$

Using Eq. (A.7) with the initial value (A.6d), one has

$$A(R) \frac{\partial T}{\partial \eta} = \pm \dot{r}(\tau) \quad (\text{A.9})$$

for all η . Using this result to substitute for $\partial T / \partial \eta$ in Eq. (A.2), one finds a first integral for Eq. (A.8):

$$\left(\frac{\partial R}{\partial \eta} \right)^2 = A(R) + \dot{r}^2(\tau) . \quad (\text{A.10})$$

Taking the square root, using the special case $\eta = 0$ (Eq. (A.6b)) to resolve the sign ambiguity, one finds

$$\frac{\partial R}{\partial \eta} = \beta(R, \dot{r}(\tau)) . \quad (\text{A.11})$$

The solution $R(\tau, \eta)$ is then determined implicitly by the integral

$$\eta = \int_{r(\tau)}^{R(\tau, \eta)} \frac{d\tilde{r}}{\beta(\tilde{r}, \dot{r}(\tau))} . \quad (\text{A.12})$$

To lowest order

$$R(\tau, \eta) = r(\tau) + \beta\eta + \mathcal{O}(\eta^2) , \quad (\text{A.13})$$

which leads to

$$K_{\theta\theta}(\tau, \eta=0) = \frac{1}{2} \frac{\partial g_{\theta\theta}}{\partial \eta} \Big|_{\eta=0} = \frac{1}{2} \frac{\partial}{\partial \eta} (R^2) \Big|_{\eta=0} = r\beta . \quad (\text{A.14})$$

Similarly, if we integrate Eq. (A.9) we obtain

$$T(\tau, \eta) = \pm \dot{r}(\tau) \int_0^\eta \frac{d\eta}{A(R(\tau, \eta))} + T(\tau, 0) , \quad (\text{A.15})$$

which to lowest order in η becomes

$$T(\tau, \eta) = T(\tau, 0) \pm \dot{\tau} \eta / A + \mathcal{O}(\eta^2) . \quad (\text{A.16})$$

Using (A.13) and (A.16) in Eq. (A.4), one obtains

$$K_{\tau\tau}(\tau, \eta=0) = \frac{1}{2} \frac{\partial g_{\tau\tau}}{\partial \eta} \Big|_{\eta=0} = -\frac{1}{\beta} \left(\dot{\tau} + \frac{1}{2} \frac{dA}{d\tau} \right) . \quad (\text{A.17})$$

APPENDIX B MATTER ACTION AT THE WALL

Here we show that the expression used in the text for the action of a thin wall can be functionally differentiated to give the correct energy-momentum tensor.

In the text, the action for a thin wall is specified by Eq. (3.6) to be $-\sigma$ times the three-volume of the world hypersurface swept out by the motion of the wall:

$$I_{\text{matter}}^{\text{wall}} = -\sigma \int d^3\xi \sqrt{q} , \quad (\text{B.1})$$

where σ is the energy per area of the wall, $q_{ij}(\xi)$ is metric within the wall, and $q \equiv -\det q_{ij}$. We will extract the energy-momentum tensor from this action by varying the spacetime metric $g_{\mu\nu}$, using the fact that $T_{\mu\nu}$ is defined by

$$\delta I = \frac{1}{2} \int d^4x \sqrt{g} T^{\mu\nu}(x) \delta g_{\mu\nu}(x) . \quad (\text{B.2})$$

Let the wall be described by the embedding in spacetime $x^\mu = \bar{X}^\mu(\xi^i)$. The metric on the wall is then given by

$$q_{ij}(\xi) = \frac{\partial \bar{X}^\mu}{\partial \xi^i} \frac{\partial \bar{X}^\nu}{\partial \xi^j} g_{\mu\nu}(\bar{X}(\xi)) . \quad (\text{B.3})$$

Varying $g_{\mu\nu}$ in (B.1) then gives

$$\begin{aligned} \delta I_{\text{matter}}^{\text{wall}} &= -\frac{1}{2} \sigma \int d^3\xi \sqrt{q} q^{ij} \frac{\partial \bar{X}^\mu}{\partial \xi^i} \frac{\partial \bar{X}^\nu}{\partial \xi^j} \delta g_{\mu\nu}(\bar{X}(\xi)) \\ &= -\frac{1}{2} \sigma \int d^4x d^3\xi \sqrt{q} q^{ij} \frac{\partial \bar{X}^\mu}{\partial \xi^i} \frac{\partial \bar{X}^\nu}{\partial \xi^j} \delta^4(x - \bar{X}(\xi)) \delta g_{\mu\nu}(x) , \end{aligned} \quad (\text{B.4})$$

where q^{ij} is the inverse of the matrix q_{ij} . Comparing (B.4) with (B.2), we arrive at

$$T^{\mu\nu} = -\sigma \int \frac{d^3\xi \sqrt{q}}{\sqrt{g}} q^{ij} \frac{\partial \bar{X}^\mu}{\partial \xi^i} \frac{\partial \bar{X}^\nu}{\partial \xi^j} \delta^4(x - \bar{X}(\xi)) . \quad (\text{B.5})$$

We now choose Gaussian normal coordinates (ξ^i, η) for the four-dimensional spacetime, so the four-metric at the wall takes the form

$$ds^2 = d\eta^2 + q_{ij}(\xi, \eta) d\xi^i d\xi^j . \quad (\text{B.6})$$

Eq. (B.5) then simplifies to

$$T^{\mu\nu} = -\sigma h^{\mu\nu} \delta(\eta) , \quad (\text{B.7})$$

where

$$\begin{aligned} h^{ij} &= q^{ij} , \\ h^{\eta\eta} &= h^{\eta i} = 0 . \end{aligned} \quad (\text{B.8})$$

This is the standard form of the energy-momentum tensor for a thin bubble wall, as given for example by Eqs. (2.9) and (3.17) of Ref. [6].

APPENDIX C

LAGRANGIANS LINEAR IN SECOND TIME DERIVATIVES

Given any Lagrangian $L(x, \dot{x}, \ddot{x})$ that depends on $\ddot{x}$ only through a linear term $g(x, \dot{x})\ddot{x}$, it is possible to rewrite L as the sum of a term depending only on x and $\dot{x}$, plus a dynamically irrelevant surface term. To do this, introduce a function $h(x, \dot{x})$ such that

$$\frac{\partial h}{\partial \dot{x}} = -g(x, \dot{x}), \quad (\text{C.1})$$

from which it follows that

$$g(x, \dot{x})\ddot{x} = \left(\frac{\partial h}{\partial x} \right) \dot{x} - \dot{h}. \quad (\text{C.2})$$

The first term contains no derivatives higher than $\dot{x}$ and the $\dot{h}$ term contributes only at the surface, so our goal has been achieved.

For the Lagrangian L_0 given by Eq. (3.17),

$$g(r, \dot{r}) = \frac{r^2}{2G} \frac{1}{\beta} \Big|_S^D, \quad (\text{C.3})$$

so that

$$h(r, \dot{r}) = -\frac{r^2}{2G} \int d\dot{r} \frac{1}{\beta} \Big|_S^D = -\frac{r^2}{2G} \tanh^{-1} \left(\frac{\dot{r}}{\beta} \right) \Big|_S^D. \quad (\text{C.4})$$

Since $-\dot{h}$ is a total derivative, it may be dropped from the Lagrangian without dynamical consequences. The dropping of this term is equivalent to the addition of the surface term, Eq. (3.32), arising from the intersection of the wall with the initial and final surfaces. The (dynamically irrelevant) term arising from the surface at spatial infinity, given by Eq. (3.38), does not appear in this approach.

APPENDIX D

OBTAINING EQUATIONS OF MOTION FROM THE ACTION

In this appendix we show how to vary the action (3.39) to obtain the equation of motion for the bubble wall. The calculation is complicated by the fact that the action is expressed as an integral over the proper time τ measured at the bubble wall, while the variation must keep the Schwarzschild time interval fixed.

One method to proceed would be to rewrite the action as an integral over Schwarzschild time, as is done in Eqs. (3.43) and (3.47). We first show, however, how to directly vary the action (3.39). Thus, we deal with an action of the form

$$I = \int_{\tau_i}^{\tau_f} L d\tau, \quad (\text{D.1})$$

where the final time τ_f is not fixed, but instead is allowed to vary so that the Schwarzschild time interval is fixed. Using Eq. (3.3), we write the Schwarzschild time interval as

$$t_S^f - t_S^i = \int_{\tau_i}^{\tau_f} F(r, \dot{r}) d\tau, \quad (\text{D.2})$$

where, in our case,

$$F(r, \dot{r}) = \beta_S / A_S. \quad (\text{D.3})$$

Varying the action (D.1) yields the usual expression plus two additional terms:

$$\delta I = \int_{\tau_i}^{\tau_f} \left\{ \frac{\partial L}{\partial r} - \frac{d}{d\tau} \left(\frac{\partial L}{\partial \dot{r}} \right) \right\} \delta r(\tau) d\tau + L(\tau_f) \delta \tau_f + \left(\frac{\partial L}{\partial \dot{r}} \right) \Big|_f \delta r_f, \quad (\text{D.4})$$

where $\delta \tau_f$ is the variation of τ_f , and δr_f is the variation of $r(\tau)$ at proper time τ_f . To determine $\delta \tau_f$, we take the variation of Eq. (D.2):

$$0 = \int_{\tau_i}^{\tau_f} \left\{ \frac{\partial F}{\partial r} - \frac{d}{d\tau} \left(\frac{\partial F}{\partial \dot{r}} \right) \right\} \delta r(\tau) d\tau + F(\tau_f) \delta \tau_f + \left(\frac{\partial F}{\partial \dot{r}} \right) \Big|_f \delta r_f. \quad (\text{D.5})$$

To solve these equations we must relate δr_f to the other quantities, which is done by noting that the value of r at T_S^f is fixed. Since the proper time corresponding to T_S^f changes from τ_f to $\tau_f + \delta \tau_f$, one has $r(\tau_f + \delta \tau_f) + \delta r(\tau_f + \delta \tau_f) = r(\tau_f)$, which linearizes to give

$$\delta r_f = -\dot{r}(\tau_f) \delta \tau_f. \quad (\text{D.6})$$

Substituting this expression into Eq. (D.5) gives

$$\delta \tau_f = - \frac{\int_{\tau_i}^{\tau_f} \left\{ \frac{\partial F}{\partial r} - \frac{d}{d\tau} \left(\frac{\partial F}{\partial \dot{r}} \right) \right\} \delta r(\tau) d\tau}{F(\tau_f) - \left(\dot{r} \frac{\partial F}{\partial \dot{r}} \right) \Big|_f}, \quad (\text{D.7})$$

which can be used with (D.4) to give

$$\delta I = \int_{\tau_i}^{\tau_f} \left\{ \frac{\partial L}{\partial \tau} - \frac{d}{d\tau} \left(\frac{\partial L}{\partial \dot{\tau}} \right) + E_f \left[\frac{\partial F}{\partial \tau} - \frac{d}{d\tau} \left(\frac{\partial F}{\partial \dot{\tau}} \right) \right] \right\} \delta \tau(\tau) d\tau, \quad (\text{D.8})$$

where

$$E_f \equiv E(\tau_f) \quad (\text{D.9})$$

and

$$E(\tau) = \frac{\dot{\tau} \frac{\partial L}{\partial \dot{\tau}} - L}{F - \dot{\tau} \frac{\partial F}{\partial \dot{\tau}}}. \quad (\text{D.10})$$

The equation of motion is then

$$\frac{\partial L}{\partial \tau} - \frac{d}{d\tau} \left(\frac{\partial L}{\partial \dot{\tau}} \right) + E_f \left[\frac{\partial F}{\partial \tau} - \frac{d}{d\tau} \left(\frac{\partial F}{\partial \dot{\tau}} \right) \right] = 0. \quad (\text{D.11})$$

One would not expect a coefficient in the equation of motion to depend on an arbitrary final time, so it is strange to see E_f in Eq. (D.11). Using Eqs. (D.10) and (D.11), however, one can show that $E(\tau)$ is conserved, so the subscript f appearing in Eq. (D.11) can be dropped.

The connection between this approach and the approach based on the Schwarzschild time Lagrangian $L_S = L/F$ of Eq. (3.47) can be seen by noting

$$\dot{\tau} = F(\tau, \dot{\tau}) r', \quad (\text{D.12})$$

where $r' \equiv dr/d\tau$. It follows that

$$\frac{\partial \dot{\tau}}{\partial r'} = \frac{F^2}{F - \dot{\tau} \frac{\partial F}{\partial \dot{\tau}}}. \quad (\text{D.13})$$

It is then straightforward to show that the conserved energy associated with L_S is identical to $E(\tau)$ as defined in (D.10):

$$H \equiv r' \frac{\partial L_S}{\partial r'} - L_S = E(\tau). \quad (\text{D.14})$$

Using the Lagrangian (3.40) for L and Eq. (D.3) for F , the conserved energy (D.10) can be evaluated as

$$E = M - \frac{r\beta_S}{G} [\beta_D - \beta_S - \kappa r]. \quad (\text{D.15})$$

For the case $E = M$ this reproduces the equation of motion (2.11). For other values of E the Lagrangian produces unphysical solutions, corresponding to the fact that our truncation of the problem to a single degree of freedom has led to the loss of the energy constraint equation.

APPENDIX E

EUCLIDEAN SCHWARZSCHILD AND DE SITTER SPACE

The Euclidean section of Schwarzschild space can be obtained by defining a Euclidean time variable $T_{SE} \equiv i\tau$, and then analytically continuing the metric to real values of T_{SE} . One obtains

$$ds^2 = A_S dT_{SE}^2 + A_S^{-1} dR^2 + R^2 d\Omega^2, \quad (\text{E.1})$$

where A_S is given by Eq. (2.2). Similarly we can introduce a Euclidean Kruskal time $V_{SE} \equiv iT_S$. The Euclidean Kruskal coordinates V_{SE} and $U_{SE} \equiv U_S$ are related to their Schwarzschild counterparts (T_{SE}, R) by analytic continuations of Eqs. (2.4) and (2.5):

$$\left(\frac{R}{2GM} - 1 \right) \exp \left(\frac{R}{2GM} \right) = U_{SE}^2 + V_{SE}^2, \quad (\text{E.2})$$

$$T_{SE} = 4GM \tan^{-1}(V_{SE}/U_{SE}). \quad (\text{E.3})$$

We see from Eq. (E.2) that in the Euclidean manifold, $R \geq 2GM$. This corresponds to the fact that the metric given by Eq. (E.1) is Euclidean only within this range. $R = 0$ is not included in the Euclidean manifold, which is therefore free of curvature singularities. It is clear from (E.3) that T_{SE} is an angular variable, running from 0 to $8\pi GM$. (In Euclidean quantum field theory this corresponds to a space with a finite Hawking temperature, $T_H = 1/8\pi GM$.) We choose the standard convention that T_{SE} is zero along the positive U_{SE} -axis.

In contrast to the Lorentzian-signature case, the Euclidean Schwarzschild coordinates cover the entire manifold. Nonetheless it is useful to describe the geometry in the analytic continuation of Kruskal coordinates. Let us introduce polar coordinates for the U_{SE} - V_{SE} plane, with $\rho_S^2 \equiv U_{SE}^2 + V_{SE}^2$ and $\Theta_S \equiv \tan^{-1} V_{SE}/U_{SE} = T_{SE}/4GM$. In these coordinates the line element takes the form

$$ds^2 = 32 \frac{(GM)^3}{R} \exp \left(-\frac{R}{2GM} \right) (d\rho_S^2 + \rho_S^2 d\Theta_S^2) + R^2 d\Omega^2. \quad (\text{E.4})$$

The geometry can be visualized as the surface of a laboratory test tube, which we describe in a vertical orientation. The point $\rho_S = 0$ corresponds to the base (i.e., the bottom-most point) of the tube, and ρ_S is a measure of distance from the base; Θ_S is the azimuthal angle. Each point on the test tube corresponds to a two-sphere of radius R . The circumference of the test tube (i.e., the proper distance traversed by varying Θ_S by 2π) is given by

$$s = 8\pi GM \sqrt{1 - \frac{2GM}{R}}, \quad (\text{E.5})$$

which increases from zero at the base (where $R = 2GM$) to a limiting value of $8\pi GM$ at large values of R .

Euclidean de Sitter space is a four-sphere of radius χ^{-1} . The simplest way to see this is to consider the embedding of de Sitter space in a five-dimensional Minkowski space.

with coordinates X^α ($\alpha = 0, 1, 2, 3, 4$) and metric $ds^2 = \eta_{\alpha\beta} dX^\alpha dX^\beta$, where $\eta_{\alpha\beta} = \text{diag}(-1, 1, 1, 1, 1)$. De Sitter space is then given by the hyperbolic subspace

$$\eta_{\alpha\beta} X^\alpha X^\beta = \chi^{-2}. \quad (\text{E.6})$$

A four-sphere of radius χ^{-1} is then obtained by analytically continuing the Minkowski time coordinate X^0 .

In static de Sitter coordinates, the analytic continuation is closely analogous to the Schwarzschild case. The metric of Euclidean de Sitter space in these coordinates is

$$ds^2 = A_D dT_{DE}^2 + A_D^{-1} dR^2 + R^2 d\Omega^2, \quad (\text{E.7})$$

where T_{DE} represents Euclidean de Sitter (static) time. A Euclidean Gibbons–Hawking time V_{DE} can be introduced, and the coordinates (V_{DE}, U_{DE}) are related to the static pair (T_{DE}, R) by the analytic continuations of (2.9) and (2.10):

$$\frac{1 - \chi R}{1 + \chi R} = U_{DE}^2 + V_{DE}^2, \quad (\text{E.8})$$

$$T_{DE} = \chi^{-1} \tan^{-1}(V_{DE}/U_{DE}). \quad (\text{E.9})$$

From Eq. (E.8) we see that $0 \leq R \leq \chi^{-1}$, and from Eq. (E.9) that T_{DE} is an angular variable running from 0 to $2\pi\chi^{-1}$ (corresponding to a Gibbons–Hawking temperature¹⁷ $T_{G-H} = \chi/2\pi$). Since the discussions in this paper emphasize the left-hand side of the Euclidean de Sitter diagram, we choose the branch of the inverse tangent in Eq. (E.9) that puts $T_{DE} = 0$ along the negative U_{DE} -axis—it is still measured positively when counter-clockwise, as usual.

Following the Schwarzschild example, we can introduce polar coordinates for the U_{DE} - V_{DE} plane, with $\rho_D^2 \equiv U_{DE}^2 + V_{DE}^2$ and $\Theta_D \equiv \tan^{-1} V_{DE}/U_{DE} = \chi T_{DE}$. The line element then takes the form

$$ds^2 = \chi^{-2} (1 + \chi R)^2 (d\rho_D^2 + \rho_D^2 d\Theta_D^2) + R^2 d\Omega^2. \quad (\text{E.10})$$

On the Euclidean Gibbons–Hawking diagram, de Sitter space is represented by the disk $\rho_D \leq 1$. Each point of the disk corresponds to a two-sphere of radius R , which degenerates to a single point on the boundary $\rho_D = 1$ ($R = 0$).

APPENDIX F DISCONTINUITIES ARISING IN THE INTEGRATION OF THE ACTION

Here we wish to show that the Euclidean Schwarzschild transit time $\bar{t}_{SE}$, defined by Eq. (5.17) as

$$\bar{t}_{SE} = \int_0^{\tau_E} \frac{\beta_{SE}}{A_S} d\tau_E, \quad (\text{F.1})$$

changes discontinuously as M passes through M_S . The discontinuity has just the right value so that $\bar{t}_{SE}^\infty$, defined by Eq. (5.28b), is continuous. There is also a discontinuity in the expression

$$\frac{4\pi}{3} \int_0^{\tau_E} r^3(\tau_E) \frac{\beta_{DE}}{A_D} d\tau_E \quad (\text{F.2})$$

as M passes through M_D . We will only evaluate the discontinuity in the first expression, however, as the analysis of the discontinuity in the second case is very similar.

We first write

$$\bar{t}_{SE} = \int_{r_1}^{r_2} \frac{\beta_{SE}}{A_S \dot{r}_E} dr, \quad (\text{F.3})$$

where $\dot{r}_E$ and β_{SE} are functions of r as a consequence of the equations of motion. $\dot{r}_E$ is positive throughout the trajectory, but vanishes at the endpoints. Its value is determined by Eq. (5.14):

$$\dot{r}_E = \sqrt{V(r, M) + 1}, \quad (\text{F.4})$$

where $V(r, M)$ is given by Eq. (2.16), which can be written as

$$V(r, M) = - \left\{ \frac{[2GM - r^3/(2GM_S)^2]^2}{4\kappa^2 r^4} + \frac{2GM}{r} \right\}, \quad (\text{F.5})$$

where

$$M_S = \frac{1}{2G\sqrt{\chi^2 + \kappa^2}}. \quad (\text{F.6})$$

β_{SE} is determined by Eq. (5.12a) as

$$\beta_{SE} = \pm \sqrt{1 - 2GM/r - \dot{r}_E^2}. \quad (\text{F.7})$$

For $M > M_S$, β_{SE} is negative throughout the trajectory, while for $M = M_S$ it vanishes at the turning point r_1 . For $M < M_S$, β_{SE} starts out positive and changes sign along the trajectory.

M_S is defined as the mass for which β_S vanishes at r_1 , so $r_1 = 2GM_S$ for $M = M_S$. $V(r, M)$ can be expanded for small $\Delta M \equiv M - M_S$ and small $\Delta r \equiv r - 2GM_S$, giving

$$V(r, M) = -1 - \frac{\Delta M}{M_S} + \frac{\Delta r}{2GM_S} + a_1 \Delta M^2 + a_2 \Delta M \Delta r + a_3 \Delta r^2 + \dots \quad (\text{F.8})$$

and

$$A_S(r, M) = -\frac{\Delta M}{M_S} + \frac{\Delta r}{2GM_S} + a_4 \Delta M \Delta r + a_5 \Delta r^2 + \dots, \quad (\text{F.9})$$

where a_i will always stand for a constant whose value does not matter.

Since the singularity of interest occurs at $r = r_1$, it is useful to change variables to $s \equiv r - r_1$:

$$\bar{t}_{SE}(M) = \int_0^{r_2 - r_1} F(s, M) ds, \quad (\text{F.10})$$

where

$$F(s, M) \equiv \frac{\sqrt{A_S - V - 1}}{A_S \sqrt{V + 1}}. \quad (\text{F.11})$$

First consider the case $M = M_S$ exactly. Then for small s we have, using $s = \Delta r$,

$$F(s, M) = \frac{\sqrt{O(s^2)}}{\left(\frac{s}{2GM_S} + \dots\right) \sqrt{\frac{s}{2GM_S} + \dots}} = O(s^{-1/2}), \quad (\text{F.12})$$

so the integral converges.

Now consider a small positive ΔM . Since $V(r_1, M) = -1$ we can use (F.8) to find $\Delta r_1 \equiv r_1 - 2GM_S$:

$$0 = -\frac{\Delta M}{M_S} + \frac{\Delta r_1}{2GM_S} + a_1 \Delta M^2 + a_2 \Delta M \Delta r_1 + a_3 \Delta r_1^2 + \dots. \quad (\text{F.13})$$

We can also expand V and A_S in s using $\Delta r = s + \Delta r_1$, finding

$$V = -1 + \frac{s}{2GM_S} + a_7 s \Delta M + a_8 s^2 + \dots, \quad (\text{F.14})$$

$$A_S = \frac{s}{2GM_S} + a_9 s \Delta M + a_{10} \Delta M^2 + a_{11} s^2 + \dots. \quad (\text{F.15})$$

Note that V has no term proportional to ΔM^2 . Now

$$F(s, M) = \frac{\sqrt{a_{12} s \Delta M + a_{10} \Delta M^2}}{\left(\frac{s}{2GM_S} + a_9 s \Delta M + a_{10} \Delta M^2 + \dots\right) \sqrt{\frac{s}{2GM_S} + \dots}}. \quad (\text{F.16})$$

Since the discontinuity of $\bar{t}_{SE}(M)$ occurs from the integration near $s = 0$, it is useful to decompose $\bar{t}_{SE}(M)$ as the sum of

$$\begin{aligned} \bar{t}_{SE}^{(1)}(M) &= \int_0^\delta F(s, M) ds \\ \bar{t}_{SE}^{(2)}(M) &= \int_\delta^{r_2 - r_1} F(s, M) ds, \end{aligned} \quad (\text{F.17})$$

where δ is a small positive number. Since the integrand for $\bar{t}_{SE}^{(2)}(M)$ has a derivative with respect to M that is bounded for all s in the range of integration, it follows that $\bar{t}_{SE}^{(2)}(M)$ is continuous in M . The desired discontinuity is then

$$\text{Disc} = \lim_{\Delta M \rightarrow 0+} \left[\bar{t}_{SE}^{(1)}(M_S + \Delta M) - \bar{t}_{SE}^{(1)}(M_S) \right]. \quad (\text{F.18})$$

Since the expression above is valid for any small $\delta > 0$, we can take the limit $\delta \rightarrow 0$, noting that $\bar{t}_{SE}^{(1)}(M_S)$ vanishes in this limit. Then

$$\text{Disc} = \lim_{\delta \rightarrow 0} \lim_{\Delta M \rightarrow 0+} \int_0^\delta \frac{\sqrt{a_{12} s \Delta M + a_{10} \Delta M^2} ds}{\left(\frac{s}{2GM_S} + a_9 s \Delta M + a_{10} \Delta M^2\right) \sqrt{\frac{s}{2GM_S}}}. \quad (\text{F.19})$$

Now let $u = s/(2GM_S a_{10} \Delta M^2)$. We see that the terms in $s \Delta M$ do not contribute, so

$$\begin{aligned} \text{Disc} &= 2GM_S \int_0^\infty \frac{du}{(u+1)\sqrt{u}} \\ &= 2\pi GM_S. \end{aligned} \quad (\text{F.20})$$

Similarly we have that $\lim_{\Delta M \rightarrow 0+} [\bar{t}_{SE}(M_S - \Delta M) - \bar{t}_{SE}(M_S)] = -2\pi GM_S$, so the net discontinuity is $4\pi GM_S$, as claimed in the text.